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To cite this article: Michael Kupper, Max Nendel & Alessandro Sgarabottolo (2025) Risk measures based on weak optimal transport, Quantitative Finance, 25:2, 163-180, DOI: [10.1080/14697688.2024.2403540](https://doi.org/10.1080/14697688.2024.2403540)

To link to this article: <https://doi.org/10.1080/14697688.2024.2403540>



Published online: 02 Oct 2024.



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Risk measures based on weak optimal transport

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(Received 21 December 2023; accepted 29 August 2024)

In this paper, we study convex risk measures with weak optimal transport penalties. In a first step, we show that these risk measures allow for an explicit representation via a nonlinear transform of the loss function. In a second step, we discuss computational aspects related to the nonlinear transform as well as approximations of the risk measures using, for example, neural networks. Our setup comprises a variety of examples, such as classical optimal transport penalties, parametric families of models, divergence risk measures, uncertainty on path spaces, moment constraints, and martingale constraints. In a last step, we show how to use the theoretical results for the numerical computation of worst-case losses in an insurance context and no-arbitrage prices of European contingent claims after quoted maturities in a model-free setting.

Keywords: Risk measure; Weak optimal transport; Neural network; Model uncertainty; Martingale optimal transport

JEL Classifications: C63, C45, C61

1. Introduction

A key component of financial modeling lies in the description of the distribution of underlying risk factors. Since these risk factors typically take values in high-dimensional vector spaces and exhibit complex dependence structures, there is a natural trade-off between the simplicity or implementability of a probabilistic model on the one hand and the most accurate description of reality on the other. Since a perfect reflection of reality in a single probabilistic framework is almost never possible, financial modeling is almost always accompanied by model uncertainty. A mathematical framework that allows to include model uncertainty in the assessment of financial contracts is given by the theory of risk measures, cf. Föllmer and Schied (2016). Under appropriate continuity conditions, every convex risk measure admits a robust representation $\rho(f) = \sup_{\nu} \int f d\nu - \alpha(\nu)$, where the evaluation $\int f d\nu$ of a financial position (or loss function) f is performed among a family of probability measures, which are penalized according to their plausibility using a suitable penalty function α .

In the past decade, a strand of literature has formed around risk measures, where the penalty term $\alpha(\nu)$ is given as a minimal cost of transportation from a given reference measure μ , cf. Bartl *et al.* (2020), or by a suitable function applied to the Wasserstein distance to the reference measure μ , see Bartl *et al.* (2021) and Nendel and Sgarabottolo

(2022), Fuhrmann *et al.* (2023) for a dynamic setting. Here, the reference measure μ might be, for example, a model that is particularly attractive due to its computational simplicity, suggested by some expert, or purely data-driven, e.g. an empirical distribution derived from given data points, cf. Obłój and Wiesel (2021). A similar idea is inherent to the discipline of distributionally robust optimization, where an additional optimization over (possibly rescaled versions of) such risk measures is performed, cf. Pflug and Wozabal (2007), Mohajerin Esfahani and Kuhn (2018), Blanchet and Murthy (2019), and Gao and Kleywegt (2023). We also refer to Bartl *et al.* (2021, 2023), and Bartl and Wiesel (2023) for an analysis of the sensitivity of robust optimization problems with respect to the degree of uncertainty. Moreover, robust versions of risk measures based on uncertainty sets are investigated in Pesenti *et al.* (2020) and Bernard *et al.* (2022).

The classical formulation of an optimal transport problem is due to Monge, where a transportation plan is a deterministic map from one state space to another. Since this problem is ill-posed, for example, when transporting a Dirac measure to an absolutely continuous measure, Kantorovich introduced a relaxation based on the notion of couplings, cf. Villani (2009) for the details. Using kernels instead of couplings, Kantorovich's problem of optimally transporting a probability measure μ on a Borel space X to another probability ν on a Borel space Y with a measurable cost function $c: X \times Y \rightarrow [0, \infty]$ can be reformulated as a weak optimal

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transport problem

$$\inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C(x, \kappa(x)) \mu(dx) \quad (1)$$

with linear weak optimal transport cost $C(x, \kappa(x)) := \int_Y c(x, y) \kappa(x, dy)$. While Kantorovich's formulation allows to spread mass according to an average cost on the product space, the weak optimal transport relaxation, based on kernels, allows to assign a different cost for spreading mass at each point.

The weak optimal transport problem was initially introduced by Gozlan *et al.* (2017) and shortly thereafter studied by Alibert *et al.* (2019). For an overview on basic results of optimal transport theory under weak transportation costs and a collection of applications from different areas, we refer to Backhoff-Veraguas and Pammer (2022). For instance, in stochastic optimization and mathematical finance, weak optimal transport appears in the context of martingale optimal transport problems, cf. Beiglböck *et al.* (2013), causal transport problems, cf. Backhoff *et al.* (2017), semimartingale transportation problems, cf. Tan and Touzi (2013), and stability results for pricing and hedging, cf. Backhoff-Veraguas *et al.* (2020).

In this article, we focus on risk measures, whose penalty functions measure the distance of a probability ν on a Borel space Y to a reference probability μ on a Borel space X by means of a weak optimal transport cost of the form (1) with a general measurable weak optimal transport cost $C: X \times \mathcal{P}(Y) \rightarrow [0, \infty]$, i.e. we consider risk measures of the form

$$\rho(f) := \sup_{\nu \in \mathcal{P}(Y)} \left(\int_Y f(y) \nu(dy) - \inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C(x, \kappa(x)) \mu(dx) \right). \quad (2)$$

The special case $C(x, \kappa(x)) = \int_Y c(x, y) \kappa(x, dy)$ with a measurable cost function $c: X \times Y \rightarrow [0, \infty]$, resulting in an optimal transport penalty $d_c(\mu, \nu) := \inf_{\pi} \int_{X \times Y} c(x, y) \pi(dx, dy)$, where the infimum is taken over all couplings π between μ and ν , has been studied in Bartl *et al.* (2020) and Nendel and Sgarabottolo (2022). Passing to the weak optimal transport cost allows to subsume a wide spectrum of examples in our framework such as

- model-free option pricing based on a finite number of quoted options, cf. section 4.7,
- model-free price bounds for European options after the last quoted maturity, cf. section 5.2,
- drift uncertainty for path-dependent options in the spirit of Föllmer (2022), cf. section 4.4,
- general risk measures on L^∞ such as expected shortfall and penalties based on φ -divergences, cf. section 4.2,
- martingale optimal transport penalties with càdlàg martingales interpolating an initial distribution μ and a terminal marginal ν , cf. section 4.6,
- parametric forms of uncertainty, cf. section 4.1.

We refer to section 4 for the details and section 5 for numerical schemes based on our theoretical results.

In a first step, we show that the risk measure ρ , given in equation (2), can be expressed in terms of a nonlinear transform of the loss function f which is independent of the reference measure μ . Theorem 2.1 shows that

$$\rho(f) = \int_X f^C(x) \bar{\mu}(dx),$$

where $f^C(x) = \sup_{\vartheta \in \mathcal{P}(Y)} (\int f d\vartheta - C(x, \vartheta))$ is the so-called C -transform of f and $\bar{\mu}$ is the extension of μ to the universal σ -algebra on X . Similar results for transport penalties of the form $\varphi(d_c(\mu, \nu))$ are given in Bartl *et al.* (2020) for convex increasing functions $\varphi: [0, \infty] \rightarrow [0, \infty]$ and in Mohajerin Esfahani and Kuhn (2018), Zhao and Guan (2018), and Gao and Kleywegt (2023) for functions of the form $\varphi = \infty \cdot \mathbb{1}_{(r, \infty)}$, which correspond to penalties taking into account all distributions in a Wasserstein ball with radius $r \geq 0$. The latter are of particular interest in the field of distributionally robust optimization. While the previous results rely on duality arguments, the case $\varphi = \text{id}$, considered in the present paper, can be treated directly using selection arguments. The employed selection arguments work for general weak optimal transport penalties on general state spaces. A dual variant of theorem 2.1, based on duality for weak optimal transport, is given in Backhoff-Veraguas *et al.* (2019, Theorem 1.3), see also Gozlan *et al.* (2017, Theorem 9.6) for an earlier duality result.

In a second step, we exploit the fact that the inner expression in the C -transform is measure affine and show that, in special yet relevant cases, the computation of the C -transform can be reduced to the extreme points. In particular, we use a classical representation of the extreme points of sets of probability measures given in terms of generalized moment constraints, cf. von Weizsäcker and Winkler (1979) and Winkler (1988). This allows also for a variational formulation of $\rho(f)$ based on an optimization over Borel measurable functions taking values in the set of extreme points, see proposition 3.6.

In the case of a Wasserstein or martingale Wasserstein penalty, this allows to use a neural network approximation in the spirit of Nendel and Sgarabottolo (2022). The idea of using neural network approximations for the computation of risk measures based on classical optimal transport has recently been exploited in Eckstein *et al.* (2020), Eckstein and Kupper (2021), see also Henry-Labordere (2019) and De Gennaro Aquino and Eckstein (2020) and the references therein for min-max methods. However, the approach contained in these works relies on duality results, which transform the original problem into the dual superhedging problem. In our approach, we use a neural network to model relevant transformations of the reference measure which are able to approximate the worst-case measure. Therefore, while the approach of Eckstein *et al.* (2020) and Eckstein and Kupper (2021) provides an approximation from above and gives as a byproduct the optimal superhedging strategy for the robust hedging problem, our approach provides an approximation from below and gives as a byproduct the C -transform of the loss function. Note that, having the C -transform of the loss function at hand, allows for a fast re-evaluation of the risk measure as the reference model changes at the cost of a Monte Carlo simulation.

This makes the approach also appealing for risk management, where the possibility to promptly adjust the evaluation of a risk as the baseline model changes is of great advantage.

An elementary reformulation of the martingale constraint, which is a priori an infinite-dimensional constraint, allows to understand it as a single moment constraint and allows us to consider robust price bounds under no-arbitrage considerations in a model-free setting, see sections 4.5 and 4.7. In section 5.2, we consider European claims with maturities that exceed the last quoted maturity and model the absence of information after the quoted maturities by penalizing deviations from the last quoted distribution of the underlying assets with a martingale Wasserstein distance. Using the neural network approximation scheme, we compute no-arbitrage bounds for the prices of options written on a possibly large number of underlying assets. An advantage of this approach, where the optimization is based on sampling from the reference distribution, is that the problem remains numerically feasible even when the number of assets grows, avoiding the classical curse of dimensionality that naturally arises in schemes that are based on a discretization of the domain. This aspect is explored in section 5.2.3.

The rest of the paper is organized as follows. Section 2 introduces the setup and contains the representation of the risk measure ρ via the C -transform of the loss function f , see theorem 2.1. In section 3, we introduce a specific class of cost functions that allow for more explicit solutions, when the set of extreme points of the ambiguity set is known, see theorem 3.3. In section 4, we present a variety of examples that are covered by our approach and indicate possible applications. Finally, in section 5, we formulate our neural network approximation based on a general approximation result for Wasserstein penalties, see proposition 5.1. This approximation based on neural networks is then applied to several examples from finance and insurance.

1.1. Notation

Given a topological space Ω , we denote by 2^Ω its power set, by $\mathcal{B}(\Omega)$ its Borel σ -algebra, and by $\mathcal{P}(\Omega)$ the set of all probability measures on $\mathcal{B}(\Omega)$, endowed with the weak topology. Let $\overline{\mathbb{R}} := \mathbb{R} \cup \{\pm\infty\}$. Then, we denote by $B(\Omega)$ the space of all Borel measurable functions $\Omega \rightarrow \overline{\mathbb{R}}$ and by $B_b(\Omega)$ the space of all bounded Borel measurable functions $\Omega \rightarrow \mathbb{R}$. For a probability measure $\mu \in \mathcal{P}(\Omega)$, let $\overline{\sigma}(\mu) := \mathcal{B}(\Omega) \vee \mathcal{N}_\mu$, where $\mathcal{N}_\mu$ consists of all sets $N \in 2^\Omega$ with $N \subset B$ for some $B \in \mathcal{B}(\Omega)$ with $\mu(B) = 0$. Then, we denote by $\mathcal{U}(\Omega) := \bigcap_{\mu \in \mathcal{P}(\Omega)} \overline{\sigma}(\mu)$ the σ -algebra of all universally measurable sets, and by $\overline{\mu}$ the extension of μ to $\mathcal{U}(\Omega)$ for all $\mu \in \mathcal{P}(\Omega)$. For a universally measurable function $f: \Omega \rightarrow \overline{\mathbb{R}}$ and $\mu \in \mathcal{P}(\Omega)$, we define

$$\begin{aligned} \overline{\mu}f &:= \int_{\Omega} f(x) \overline{\mu}(dx) := \int_{\Omega} f^+(x) \overline{\mu}(dx) \\ &\quad - \int_{\Omega} f^-(x) \overline{\mu}(dx) \in \overline{\mathbb{R}}. \end{aligned}$$

Here and throughout, we use the convention $\infty - \infty = -\infty$. If $f \in B(\Omega)$, we simply write μf and $\overline{\mu}f$ instead of $\int_{\Omega} f(x) \mu(dx)$ and $\int_{\Omega} f(x) \overline{\mu}(dx)$, respectively. For $\mu \in$

$\mathcal{P}(\Omega)$, we denote by $\mathcal{L}^1(\mu)$ the space of all $f \in B(\Omega)$ with $\int_{\Omega} |f(x)| \mu(dx) < \infty$, where we use the convention $|\infty| = \infty$. In a similar way, we denote by $\mathcal{L}^1(\overline{\mu})$ the space of all universally measurable functions $f: \Omega \rightarrow \overline{\mathbb{R}}$ with $\int_{\Omega} |f(x)| \overline{\mu}(dx) < \infty$.

Let $(S, \mathcal{S})$ be a measurable space. We denote the space of all $\mathcal{B}(\Omega)$ - $\mathcal{S}$ -measurable functions $\Omega \rightarrow S$ by $B(\Omega, S)$. For a probability measure $\mu \in \mathcal{P}(\Omega)$ and a function $g \in B(\Omega, S)$, we define the push-forward $g(\mu): \mathcal{S} \rightarrow [0, 1]$ by

$$(g(\mu))(A) := \mu(g \in A) \quad \text{for all } A \in \mathcal{S}.$$

2. Weak transport penalties

Throughout, let X, Y be two Borel spaces, cf. Bertsekas and Shreve (1978, Definition 7.7). Since Y is a Borel space, the set $\mathcal{P}(Y)$ of all Borel probability measures on Y is again a Borel space, cf. Bertsekas and Shreve (1978, Corollary 7.25.1). We denote by $\text{Ker}(X, Y)$ the set of all Borel measurable stochastic kernels κ from X to Y , i.e. $\kappa: X \times \mathcal{B}(Y) \rightarrow [0, 1]$ with

- (i) $\kappa(x) = \kappa(x, \cdot) \in \mathcal{P}(Y)$ for all $x \in X$,
- (ii) the map $X \rightarrow [0, 1]$, $x \mapsto \kappa(x, B)$ is Borel measurable for all $B \in \mathcal{B}(Y)$.

For $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$, we define

$$\text{Ker}(\mu, \nu) := \{\kappa \in \text{Ker}(X, Y) : \mu\kappa = \nu\},$$

where $(\mu\kappa)(B) := \int_X \kappa(x, B) \mu(dx)$ for all $B \in \mathcal{B}(Y)$.

In the following, let $\mu \in \mathcal{P}(X)$ be a reference measure and $C: X \times \mathcal{P}(Y) \rightarrow [0, \infty]$ be a Borel measurable cost function. We assume that there exists some $\kappa_0 \in \text{Ker}(X, Y)$ with $C(x, \kappa_0(x)) = 0$ for μ -almost all $x \in X$. For $f \in B(Y)$, let

$$\begin{aligned} \rho(f) &:= \sup_{\nu \in \mathcal{P}(Y)} \left(\int_Y f(y) \nu(dy) \right. \\ &\quad \left. - \inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C(x, \kappa(x)) \mu(dx) \right), \end{aligned} \quad (3)$$

where we use the convention $\inf \emptyset = \infty$. By assumption,

$$\int_Y f(y) (\mu\kappa_0)(dy) \leq \rho(f) \quad \text{for all } f \in B(Y),$$

i.e. the kernel κ_0 can be interpreted as a transform of states in X into states in Y , which is not penalized, or, in other words, ρ incorporates a risk loading w.r.t. the transformed reference measure $\mu\kappa_0$. Observe that the functional $\rho: B(Y) \rightarrow \overline{\mathbb{R}}$ defines a *monetary risk measure*, i.e.

- (i) $\rho(f) \leq \rho(g)$ for all $f, g \in B(Y)$ with $f \leq g$,[†]
- (ii) $\rho(0) = 0$ and $\rho(f + m) = \rho(f) + m$ for all $f \in B(Y)$ and $m \in \mathbb{R}$,

and that the weak optimal transportation cost $\alpha(\nu) := \inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C(x, \kappa(x)) \mu(dx)$ for each $\nu \in \mathcal{P}(Y)$ is a penalty function of the risk measure ρ . We refer to Föllmer

[†] For $f, g \in B(X)$, we write $f \leq g$ if $f(x) \leq g(x)$ for all $x \in X$.

and Schied (2016, Chapter 4) for a detailed discussion on monetary risk measures.

We define the C -transform of $f \in B(Y)$ by

$$f^C(x) := \sup_{\vartheta \in \mathcal{P}(Y)} (\vartheta f - C(x, \vartheta)) \in \overline{\mathbb{R}} \quad \text{for all } x \in X.$$

Since $C(x, \kappa_0(x)) = 0$ for μ -almost all $x \in X$, it follows that $\kappa_0(x)f \leq f^C(x)$ for μ -almost all $x \in X$. Moreover, f^C is upper semianalytic and therefore universally measurable. In fact, by Bertsekas and Shreve (1978, Corollary 7.29.1), for $f \in B(Y)$, the map $\vartheta \mapsto \vartheta f$ is Borel measurable, which implies that also the map $(x, \vartheta) \mapsto \vartheta f - C(x, \vartheta)$ is Borel measurable. Then, since Borel measurable maps are upper semianalytic, by Bertsekas and Shreve (1978, Proposition 7.47) we obtain that f^C is upper semianalytic for all $f \in B(Y)$. In the special case, where $C(x, \delta_y) := c(x, y)$ for all $x \in X$ and $y \in Y$ with some Borel measurable function $c: X \times Y \rightarrow [0, \infty]$, and $C(x, \vartheta) := \infty$ for all $x \in X$ if $\vartheta \in \mathcal{P}(Y)$ is not a Dirac measure, the C -transform reduces to the well-known c -transform $f^c(x) = \sup_{y \in Y} (f(y) - c(x, y))$ for all $x \in X$.

In the following, we restrict the risk measure ρ to the set

$$\mathcal{L} := \left\{ f \in B(Y) : -\infty < \int_Y f(y) (\mu \kappa_0)(dy) \text{ and } \int_X f^C(x) \overline{\mu}(dx) < \infty \right\}.$$

Observe that $B_b(Y) \subset \mathcal{L}$. In particular, $\mathcal{L}$ contains all constant functions. The following characterization of the risk measure (3) is useful for its numerical computation in section 5.

THEOREM 2.1 *The set $\mathcal{L}$ is convex and $\rho: \mathcal{L} \rightarrow \mathbb{R}$ defines a convex risk measure, i.e.*

$$\rho(\lambda f + (1 - \lambda)g) \leq \lambda \rho(f) + (1 - \lambda)\rho(g) \\ \text{for all } f, g \in \mathcal{L} \text{ and } \lambda \in [0, 1].$$

Moreover, for all $f \in \mathcal{L}$,

$$\rho(f) = \sup_{\kappa \in \text{Ker}(X, Y)} \int_X \kappa(x)f - C(x, \kappa(x)) \mu(dx) \\ = \int_X f^C(x) \overline{\mu}(dx). \quad (4)$$

Proof Let $f \in \mathcal{L}$. Since $\inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C(x, \kappa(x)) \mu(dx) = \infty$ whenever $\text{Ker}(\mu, \nu) = \emptyset$ and $\mu\kappa = \nu$ for all $\kappa \in \text{Ker}(\mu, \nu)$, we find that

$$\rho(f) = \sup_{\nu \in \mathcal{P}(Y)} \sup_{\kappa \in \text{Ker}(\mu, \nu)} \left(\mu\kappa f - \int_X C(x, \kappa(x)) \mu(dx) \right) \\ = \sup_{\kappa \in \text{Ker}(X, Y)} \left(\mu\kappa f - \int_X C(x, \kappa(x)) \mu(dx) \right) \\ = \sup_{\kappa \in \text{Ker}(X, Y)} \int_X \kappa(x)f - C(x, \kappa(x)) \mu(dx).$$

Hence,

$$-\infty < \int_Y f(y) (\mu \kappa_0)(dy) = \mu \kappa_0 f \leq \rho(f) \quad \text{and}$$

$$\rho(f) \leq \int_X f^C(x) \overline{\mu}(dx) < \infty. \quad (5)$$

Now, let $\varepsilon > 0$ and $\vartheta: X \rightarrow \mathcal{P}(Y)$ be a universally measurable ε -selection of f^C , i.e.

$$f^C(x) \leq \vartheta(x)f - C(x, \vartheta(x)) + \varepsilon \quad \text{for all } x \in X,$$

cf. Bertsekas and Shreve (1978, Proposition 7.50). Then, by Bertsekas and Shreve (1978, Lemma 7.28), there exists some $\kappa \in \text{Ker}(X, Y)$ with $\kappa(x) = \vartheta(x)$ for μ -almost all $x \in X$. We thus obtain that

$$\begin{aligned} \int_X f^C(x) \overline{\mu}(dx) &\leq \int_X \vartheta(x)f - C(x, \vartheta(x)) \overline{\mu}(dx) + \varepsilon \\ &= \int_X \kappa(x)f - C(x, \kappa(x)) \mu(dx) \\ &\quad + \varepsilon \leq \rho(f) + \varepsilon. \end{aligned}$$

Now, (4) follows by passing to the limit $\varepsilon \downarrow 0$.

It remains to show that $\mathcal{L}$ is convex and that $\rho: \mathcal{L} \rightarrow \mathbb{R}$ is convex. To that end, let $f, g \in \mathcal{L}$ and $\lambda \in [0, 1]$. By (5), it follows that $f, g \in \mathcal{L}^1(\mu \kappa_0)$ and $f^C, g^C \in \mathcal{L}^1(\overline{\mu})$. Since $\mathcal{L}^1(\mu \kappa_0)$ and $\mathcal{L}^1(\overline{\mu})$ are vector spaces, $\lambda f + (1 - \lambda)g \in \mathcal{L}^1(\mu \kappa_0)$ and $\lambda f^C + (1 - \lambda)g^C \in \mathcal{L}^1(\overline{\mu})$. Hence, in view of (4), both the convexity of $\mathcal{L}$ and $\rho: \mathcal{L} \rightarrow \mathbb{R}$ follow as soon as we have shown that

$$(\lambda f + (1 - \lambda)g)^C \leq \lambda f^C + (1 - \lambda)g^C \quad \overline{\mu}\text{-almost surely}. \quad (6)$$

Since $f^C, g^C \in \mathcal{L}^1(\overline{\mu})$, there exists a set $A \in \mathcal{U}(X)$ with $\overline{\mu}(A) = 1$ as well as

$$f^C(x) < \infty \quad \text{and} \quad g^C(x) < \infty \quad \text{for all } x \in A.$$

Let $x \in A$ and $\vartheta \in \mathcal{P}(Y)$ with $C(x, \vartheta) < \infty$. Then, $\vartheta f < \infty$ and $\vartheta g < \infty$. Hence,

$$\begin{aligned} \vartheta(\lambda f + (1 - \lambda)g) - C(x, \vartheta) &= (\lambda \vartheta f + (1 - \lambda)\vartheta g) - C(x, \vartheta) \\ &\leq \lambda f^C(x) + (1 - \lambda)g^C(x) \end{aligned}$$

for all $x \in A$. Taking the supremum over all $\vartheta \in \mathcal{P}(Y)$ with $C(x, \vartheta) < \infty$, it follows that

$$\begin{aligned} (\lambda f + (1 - \lambda)g)^C(x) &= \sup_{\substack{\vartheta \in \mathcal{P}(Y) \\ C(x, \vartheta) < \infty}} (\vartheta(\lambda f + (1 - \lambda)g) - C(x, \vartheta)) \\ &\leq \lambda f^C(x) + (1 - \lambda)g^C(x) \end{aligned}$$

for all $x \in A$. We have therefore shown the validity of (6), and the claim follows. $\blacksquare$

3. Optimal transport costs with additional constraints

In this section, we consider weak transport costs with constraints given in terms of a convex and Borel measurable set $\mathcal{M} \subset \mathcal{P}(Y)$ and a family $(\tau_x)_{x \in X}$ of bijective maps $Y \rightarrow Y$. We assume that both $(x, y) \mapsto \tau_x(y)$ and $(x, y) \mapsto \tau_x^{-1}(y)$ are Borel measurable maps $X \times Y \rightarrow Y$. Then, by Bertsekas and Shreve (1978, Lemma 7.12 and Proposition 7.25), the maps $(x, \vartheta) \mapsto \tau_x(\vartheta)$ and $(x, \vartheta) \mapsto \tau_x^{-1}(\vartheta)$ are Borel measurable $X \times \mathcal{P}(Y) \rightarrow \mathcal{P}(Y)$. Moreover, for all $x \in X$, the map $\mathcal{P}(Y) \rightarrow \mathcal{P}(Y)$, $\vartheta \mapsto \tau_x(\vartheta)$ is bijective with inverse $\mathcal{P}(Y) \rightarrow \mathcal{P}(Y)$, $\vartheta \mapsto \tau_x^{-1}(\vartheta)$.

Throughout this section, we consider a weak optimal transport cost of the form

$$C(x, \vartheta) = \begin{cases} \int_Y c(x, y) \vartheta(dy), & \text{if } \tau_x^{-1}(\vartheta) \in \mathcal{M}, \\ \infty, & \text{otherwise,} \end{cases} \quad (7)$$

where $c: X \times Y \rightarrow [0, \infty]$ is a Borel measurable cost function. Since c , $\mathcal{M}$, and the map $X \times \mathcal{P}(Y) \rightarrow \mathcal{P}(Y)$, $(x, \vartheta) \mapsto \tau_x^{-1}(\vartheta)$ are Borel measurable, it follows that the cost function $C: X \times \mathcal{P}(Y) \rightarrow [0, \infty]$ is Borel measurable. Again, we assume that there exists some $\kappa_0 \in \text{Ker}(X, Y)$ with $C(x, \kappa_0(x)) = 0$ for μ -almost all $x \in X$, i.e. $\tau_x^{-1}(\kappa_0(x)) \in \mathcal{M}$ and $\int_Y c(x, y) \kappa_0(x, dy) = 0$ for μ -almost all $x \in X$. Since the map $\mathcal{P}(Y) \rightarrow \mathcal{P}(Y)$, $\vartheta \mapsto \tau_x(\vartheta)$ is bijective with inverse $\mathcal{P}(Y) \rightarrow \mathcal{P}(Y)$, $\vartheta \mapsto \tau_x^{-1}(\vartheta)$, the C -transform of $f \in B(Y)$ is given by

$$\begin{aligned} f^C(x) &= \sup_{\vartheta \in \mathcal{M}} (\tau_x(\vartheta)f - C(x, \tau_x(\vartheta))) \\ &= \sup_{\vartheta \in \mathcal{M}} \int_Y \tau_x(f(y) - c(x, y)) \vartheta(dy), \end{aligned}$$

where, for $g \in B(X \times Y)$, we use the notation

$$\tau_x g(x, y) := g(x, \tau_x(y)) \quad \text{for all } x \in X \text{ and } y \in Y.$$

The following remark shows how, in the unconstrained case, this setup reduces to a penalty function, which is given in terms of a classical optimal transport problem. For additional examples of penalty functions and the related risk measures covered by this setup, we refer to section 4 below.

REMARK 3.1 In the unconstrained case, $\mathcal{M} = \mathcal{P}(Y)$ and $\tau_x = \text{id}_Y$ for all $x \in X$, it follows that $C(x, \vartheta) = \int_Y c(x, y) \vartheta(dy)$, and the weak optimal transport problem

$$\inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C(x, \kappa(x)) \mu(dx)$$

is equivalent to the classical optimal transport problem

$$\inf_{\pi \in \text{Cpl}(\mu, \nu)} \int_{X \times Y} c(x, y) \pi(dx, dy),$$

where $\text{Cpl}(\mu, \nu)$ is the set of couplings between μ and ν , i.e. the set of all probability measures $\pi \in \mathcal{P}(X \times Y)$ with $\text{pr}_X(\pi) = \mu$ and $\text{pr}_Y(\pi) = \nu$. In fact, on the one hand, for every $\kappa \in \text{Ker}(\mu, \nu)$, the condition $\mu\kappa = \nu$ implies

that the measure $\pi \in \mathcal{P}(X \times Y)$ given by $\pi(A \times B) := \int_A \kappa(x, B) \mu(dx)$ for all $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$ is a coupling between μ and ν . On the other hand, since X and Y are Borel spaces, for each $\pi \in \text{Cpl}(\mu, \nu)$, there exists a Borel measurable stochastic kernel $\kappa: X \times \mathcal{B}(Y) \rightarrow [0, 1]$ with $\pi(A \times B) = \int_A \kappa(x, B) \mu(dx)$ for all $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$, cf. Bertsekas and Shreve (1978, Corollary 7.27.2), which is an element of $\text{Ker}(\mu, \nu)$ since $\int_X \kappa(x, B) \mu(dx) = \pi(X \times B) = \nu(B)$ for all $B \in \mathcal{B}(Y)$.

In the following, we focus on the general case, where $\mathcal{M}$ is not necessarily $\mathcal{P}(Y)$. Recall that a probability measure $\theta \in \mathcal{M}$ is an extreme point of $\mathcal{M}$, if $\theta = \lambda\theta_1 + (1 - \lambda)\theta_2$ for $\theta_1, \theta_2 \in \mathcal{M}$ and $\lambda \in (0, 1)$ implies $\theta = \theta_1 = \theta_2$. We denote by $\text{ex } \mathcal{M}$ the set of all extreme points of $\mathcal{M}$. The set $\text{ex } \mathcal{M}$ is endowed with the smallest σ -algebra $\Sigma(\text{ex } \mathcal{M})$ on $\text{ex } \mathcal{M}$ such that $\text{ex } \mathcal{M} \rightarrow [0, 1]$, $\theta \mapsto \theta(B)$ is measurable for all $B \in \mathcal{B}(Y)$. By Bertsekas and Shreve (1978, Proposition 7.25), $\Sigma(\text{ex } \mathcal{M})$ coincides with the trace σ -algebra of $\mathcal{B}(\mathcal{P}(Y))$ on $\text{ex } \mathcal{M}$, which is the same as the Borel σ -algebra of the subspace topology of the weak topology on $\text{ex } \mathcal{M}$.

The next result relies on a characterization given in von Weizsäcker and Winkler (1979) and Winkler (1988). The set $\mathcal{M}$ is said to satisfy the *integral representation property (IRP)* if, for each $\vartheta \in \mathcal{M}$, there exists a probability measure $p: \Sigma(\text{ex } \mathcal{M}) \rightarrow [0, 1]$ such that

$$\vartheta(B) = \int_{\text{ex } \mathcal{M}} \theta(B) p(d\theta) \quad \text{for all } B \in \mathcal{B}(Y).$$

For the sake of illustration, we provide three examples for sets that satisfy the IRP and refer to von Weizsäcker and Winkler (1979) for a more profound study of this property.

EXAMPLE 3.2 (a) Let $\mathcal{M} \subset \mathcal{P}(Y)$ be nonempty, convex, and weakly closed. Then, $\mathcal{M}$ has the IRP, cf. von Weizsäcker and Winkler (1979, Corollary 1).
(b) Let $\varphi: Y \rightarrow \mathbb{R}$ be Borel measurable. Then,

$$\mathcal{M} := \{\vartheta \in \mathcal{P}(Y): \varphi \in \mathcal{L}^1(\vartheta)\}$$

satisfies the IRP, cf. von Weizsäcker and Winkler (1979, Corollary 3).

(c) For each $m \in \mathbb{N}$, let $\varphi_m: Y \rightarrow \mathbb{R}$ be Borel measurable and $a_m \in \mathbb{R}$. Then, the set

$$\mathcal{M} := \left\{ \vartheta \in \mathcal{P}(Y): \varphi_m \in \mathcal{L}^1(\vartheta) \text{ and } \int_Y \varphi_m(y) \vartheta(dy) \leq a_m \text{ for all } m \in \mathbb{N} \right\}$$

has the IRP, cf. von Weizsäcker and Winkler (1979, Corollary 3).

While, in von Weizsäcker and Winkler (1979, Corollary 3), it is shown that all sets in the previous example satisfy the IRP, an explicit representation of the set of extreme points $\text{ex } \mathcal{M}$ is, in general, not attainable. However, in the special case of finitely many generalized moment constraints, the extreme points are explicitly given in terms of a class of finitely supported measures, cf. theorem 3.5. The IRP is of

particular interest for linear optimization problems since, for every *measure affine* functional on $\mathcal{M}$, the supremum with respect to probability measures in $\mathcal{M}$ can be restricted to the supremum over the set of extreme points $\text{ex } \mathcal{M}$, see Winkler (1988, Theorem 3.2 and Proposition 3.1). In fact, for $\vartheta \in \mathcal{P}(Y)$ and every probability measure $p: \Sigma(\text{ex } \mathcal{M}) \rightarrow [0, 1]$ with $\vartheta(B) = \int_{\text{ex } \mathcal{M}} \theta(B) p(d\theta)$ for all $B \in \mathcal{B}(Y)$, the standard machine in measure theory, cf. Williams (1991, Chapter 5.12), yields the *barycentrical formula*[†]

$$\vartheta g = \int_{\text{ex } \mathcal{M}} \theta g p(d\theta) \quad \text{for all } g \in B(Y).$$

In particular, if $\mathcal{M} \subset \mathcal{P}(Y)$ satisfies the IRP and $g \in B(Y)$, then $\vartheta g \leq \sup_{\theta \in \text{ex } \mathcal{M}} \theta g$ for all $\vartheta \in \mathcal{M}$. Taking the supremum over all $\vartheta \in \mathcal{M}$, we get

$$\sup_{\vartheta \in \mathcal{M}} \int_Y g(y) \vartheta(dy) = \sup_{\theta \in \text{ex } \mathcal{M}} \int_Y g(y) \theta(dy). \quad (8)$$

In combination with theorem 2.1, we obtain the following result, which makes the computation of the C -conjugate f^C more tractable.

THEOREM 3.3 *Assume that the cost function C is of the form (7), and let $\text{ex } \mathcal{M} \subset \Theta \subset \mathcal{M}$. If $\mathcal{M}$ satisfies the IRP, then*

$$\rho(f) = \int_X \sup_{\theta \in \Theta} \left(\int_Y \tau_x(f(y) - c(x, y)) \theta(dy) \right) \bar{\mu}(dx) \\ \text{for all } f \in \mathcal{L}.$$

Proof Let $f \in \mathcal{L}$. Using equation (8), it follows that

$$\begin{aligned} f^C(x) &= \sup_{\vartheta \in \mathcal{M}} (\tau_x(\vartheta)f - C(x, \tau_x(\vartheta))) \\ &\geq \sup_{\theta \in \Theta} (\tau_x(\theta)f - C(x, \tau_x(\theta))) \\ &\geq \sup_{\vartheta \in \text{ex } \mathcal{M}} (\tau_x(\vartheta)f - C(x, \tau_x(\vartheta))) \\ &= \sup_{\vartheta \in \text{ex } \mathcal{M}} \int_Y \tau_x(f(y) - c(x, y)) \vartheta(dy) \\ &= \sup_{\vartheta \in \mathcal{M}} \int_Y \tau_x(f(y) - c(x, y)) \vartheta(dy) = f^C(x). \end{aligned}$$

Hence,

$$\begin{aligned} f^C(x) &= \sup_{\theta \in \Theta} (\tau_x(\theta)f - C(x, \tau_x(\theta))) \\ &= \sup_{\theta \in \Theta} \int_Y \tau_x(f(y) - c(x, y)) \theta(dy) \quad \text{for all } x \in X. \end{aligned}$$

Since f^C is universally measurable, the claim follows from theorem 2.1. $\blacksquare$

The following proposition exploits the explicit structure of the extreme points of a set of probability measures. Note that this result is a variant of Bartl *et al.* (2020, Theorem 2.4).

[†] Recall that $\vartheta g := \int_Y g^+(y) \vartheta(dy) - \int_Y g^-(y) \vartheta(dy)$ with the convention $\infty - \infty = -\infty$.

While the proof in Bartl *et al.* (2020) combines a version of the Choquet's capacitability theorem with duality arguments, our proof relies on theorem 2.1 and the characterization of extreme points of the set of all probability measures on a Borel space.

PROPOSITION 3.4 *Suppose that $\mathcal{M}$ contains all Dirac measures on Y and that the cost function C is of the form (7). Then,*

$$\begin{aligned} \rho(f) &= \int_X \sup_{y \in Y} (f(y) - c(x, y)) \bar{\mu}(dx) \\ &= \int_X f^c(x) \bar{\mu}(dx) \quad \text{for all } f \in \mathcal{L}. \end{aligned}$$

Proof By Bertsekas and Shreve (1978, Proposition 7.17), every Borel probability measure on a Borel space is regular. Hence, the set of all probability measures on a Borel space has the Dirac measures as extreme points, see Topsøe (1970, Theorem 11.1). Since $\mathcal{M}$ contains all the Dirac measures on Y , it follows that

$$\begin{aligned} f^C(x) &= \sup_{\vartheta \in \mathcal{M}} (\tau_x(\vartheta)f - C(x, \tau_x(\vartheta))) \\ &\geq \sup_{y \in Y} \tau_x(f(y) - c(x, y)) \\ &= \sup_{\vartheta \in \mathcal{P}(Y)} (\tau_x(\vartheta)f - C(x, \tau_x(\vartheta))) \geq f^c(x). \end{aligned}$$

We have therefore shown that

$$\begin{aligned} f^C(x) &= \sup_{y \in Y} \tau_x(f(y) - c(x, y)) \\ &= \sup_{y \in Y} (f(y) - c(x, y)) = f^c(x) \quad \text{for all } x \in X, \end{aligned}$$

where the second equality follows from the fact that $\tau_x: Y \rightarrow Y$ is bijective for all $x \in X$, and the claim follows again from theorem 2.1. $\blacksquare$

3.1. Generalized moment constraints

Theorem 3.3 is particularly useful when the set of extreme points of $\mathcal{M}$ has a simple and explicit representation. This is, for example, the case if we impose *generalized moment constraints* in the sense that we consider the set

$$\mathcal{M} = \left\{ \vartheta \in \mathcal{P}(Y) : f_i \in \mathcal{L}^1(\vartheta) \text{ and } \int_Y f_i(y) \vartheta(dy) \leq c_i \text{ for } 1 \leq i \leq n \right\} \quad (9)$$

with $n \in \mathbb{N}_0$, Borel measurable functions $f_1, \dots, f_n: Y \rightarrow \overline{\mathbb{R}}$, and $c_1, \dots, c_n \in \mathbb{R}$. As already remarked in example 3.2(c)), the set $\mathcal{M}$ satisfies the IRP. As remarked in the proof of proposition 3.4, the set of all Borel probability measures on Y has the Dirac measures as extreme points. This allows us to use the following theorem by Winkler, see Winkler (1988, Theorem 2.1), which we report here in the special case $\mathcal{P} = \mathcal{P}(Y)$.

THEOREM 3.5 (Winkler 1988) *Let $\mathcal{M} \subset \mathcal{P}(Y)$ be defined as in (9). Then, the set $\mathcal{M}$ is convex and*

$$\text{ex } \mathcal{M} \subset \left\{ \vartheta \in \mathcal{M} : \vartheta = \sum_{i=1}^m a_i \delta_{y_i} \text{ with } y_i \in Y, a_i > 0, \right. \\ \left. \sum_{i=1}^m a_i = 1, 1 \leq m \leq n+1, \right. \\ \left. \text{and the vectors } (f_1(y_i), \dots, f_n(y_i), 1), 1 \leq i \leq m, \right. \\ \left. \text{are linearly independent} \right\}.$$

Equality of sets holds if the moment conditions in (9) are given by equalities.

The following proposition, which is based on theorem 3.5, is a helpful tool for the numerical computation of the risk measure ρ using neural networks.

PROPOSITION 3.6 *Assume that the cost function C is of the form (7) with $\mathcal{M}$ given by (9). Then,*

$$\begin{aligned} \rho(f) &= \sup_{(a,y) \in \mathcal{B}(X, \Theta)} \int_X \sum_{i=1}^{n+1} a_i \tau_x(f(y_i(x)) \\ &\quad - c(x, y_i(x))) \mu(dx) \\ &= \int_X \sup_{(a,y) \in \Theta} \sum_{i=1}^{n+1} a_i \tau_x(f(y_i) - c(x, y_i)) \bar{\mu}(dx) \\ &\quad \text{for all } f \in \mathcal{L}, \end{aligned} \quad (10)$$

where

$$\Theta := \left\{ (a, y) \in [0, 1]^{n+1} \times Y^{n+1} : \sum_{i=1}^{n+1} a_i \delta_{y_i} \in \mathcal{M} \right\}.$$

Proof By theorem 3.3, it follows that

$$\begin{aligned} \rho(f) &= \int_X \sup_{(a,y) \in \Theta} \sum_{i=1}^{n+1} a_i \tau_x(f(y_i) - c(x, y_i)) \bar{\mu}(dx) \\ &\quad \text{for all } f \in \mathcal{L}. \end{aligned}$$

In particular, $\geq$ holds in (10). Since the map from $[0, 1] \times Y$ to the set of all subprobability measures on $\mathcal{B}(Y)$, given by $(a, y) \mapsto a \delta_y$, is continuous and $\mathcal{M}$ is weakly closed, it follows that $\Theta \subset [0, 1]^{n+1} \times Y^{n+1}$ is closed. Hence, by Bertsekas and Shreve (1978, Lemma 7.28 and Proposition 7.50), for all $f \in \mathcal{L}$ and $\varepsilon > 0$, there exists a measurable map $(a, y) : X \rightarrow \Theta$ with

$$\rho(f) \leq \int_X \sum_{i=1}^{n+1} a_i \tau_x(f(y_i(x)) - c(x, y_i(x))) \mu(dx) + \varepsilon.$$

Taking the supremum over all measurable maps $X \rightarrow \Theta$ and letting $\varepsilon \downarrow 0$, the claim follows. ■

While theorem 2.1 shows that the risk measure ρ can be computed by solving a pointwise optimization that is independent of the reference measure μ , proposition 3.6 suggests a variational approach, where the global behavior of the loss function f , the cost c , and the reference measure μ are considered simultaneously. Since the maximization involves an integral with respect to the reference distribution, this optimization aligns with the framework of stochastic gradient descent methods, where tools such as neural networks typically perform very well. In section 5 below, we will show how a neural network approximation can be formally justified in this context.

We conclude this section with the following corollary for a general set $\mathcal{M}$ containing all Dirac measures.

COROLLARY 3.7 *Suppose that $\mathcal{M}$ contains all Dirac measures on Y and that the cost function C is of the form (7). Then,*

$$\rho(f) = \sup_{y \in \mathcal{B}(X, Y)} \int_X f(y(x)) - c(x, y(x)) \mu(dx) \quad \text{for all } f \in \mathcal{L}.$$

Proof By proposition 3.4, it follows that

$$\rho(f) = \int_X \sup_{y \in Y} (f(y) - c(x, y)) \bar{\mu}(dy).$$

Hence, using proposition 3.6 in the unconstrained case, the claim follows. ■

4. Examples and applications

In this section, we collect several examples that clarify the use of weak transport penalties and discuss potential applications for the related risk measure.

4.1. Parametric uncertainty

In this subsection, we present a setup that allows to incorporate special forms of parametric uncertainty via a weak optimal transport penalization. To that end, let $\mu \in \mathcal{P}(X)$ and Θ be a Borel space. Moreover, we assume that there exists a Borel measurable and injective map $\gamma : \Theta \rightarrow \mathcal{P}(Y)$, $\theta \mapsto \gamma_\theta$, and define $\mathcal{M} := \gamma(\Theta) = \{\gamma_\theta : \theta \in \Theta\}$. Since $\mathcal{P}(Y)$ is a Borel space, it is separable and metrizable. Hence, by Kuratowski's theorem, cf. Bertsekas and Shreve (1978, Proposition 7.15), it follows that $\mathcal{M}$ is Borel measurable and that the inverse $\gamma^{-1} : \mathcal{M} \rightarrow \Theta$ is Borel measurable. We consider a Borel measurable cost function $c : X \times \Theta \rightarrow [0, \infty]$, and assume that there exists a measurable map $\theta_0 : X \rightarrow \Theta$ such that $c(x, \theta_0(x)) = 0$ for μ -almost all $x \in X$. For $x \in X$ and $\vartheta \in \mathcal{P}(Y)$, we define

$$C(x, \vartheta) := \begin{cases} c(x, \gamma^{-1}(\vartheta)), & \text{if } \vartheta \in \mathcal{M}, \\ \infty, & \text{otherwise.} \end{cases}$$

Since c , $\mathcal{M}$, and γ^{-1} are Borel measurable, it follows that $C : X \times \mathcal{P}(Y) \rightarrow [0, \infty]$ is Borel measurable. Then, by

theorem 2.1,

$$\begin{aligned}\rho(f) &= \int_X \sup_{\theta \in \Theta} (\gamma_\theta f - c(x, \theta)) \bar{\mu}(dx) \\ &= \sup_{\theta \in B(X, \Theta)} \int_X (\gamma_{\theta(x)} f - c(x, \theta(x))) \mu(dx) \\ &\text{for all } f \in \mathcal{L}.\end{aligned}$$

EXAMPLE 4.1 Let $X = Y = \mathbb{R}$, $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and $\xi: \Omega \rightarrow \mathbb{R}$ be a random variable with $\mathbb{E}_{\mathbb{P}}(\xi) = 0$. For $\theta = (m, \sigma) \in \Theta := \mathbb{R} \times [0, \infty)$, let

$$\gamma_\theta := \mathbb{P} \circ (m + \sigma \xi)^{-1}.$$

Then, the map $\gamma: \Theta \rightarrow \mathcal{P}(\mathbb{R})$, $\theta \mapsto \gamma_\theta$ is continuous, and we define the cost

$$\begin{aligned}c(x, \theta) &:= \varphi(|m - x|^2) + \psi(|\sigma|^2) \\ &\text{for } x \in X \text{ and } \theta = (m, \sigma) \in \Theta\end{aligned}$$

with nondecreasing cost functions $\varphi, \psi: [0, \infty) \rightarrow [0, \infty]$ with $\varphi(0) = \psi(0) = 0$. Defining $\theta_0(x) := (x, 0)$ for all $x \in \mathbb{R}$, it follows that $c(x, \theta_0(x)) = 0$ for all $x \in \mathbb{R}$.

4.2. Dominated risk measures, φ -divergences, and expected shortfall

Let $X = Y$, $\mu \in \mathcal{P}(X)$, and

$$\mathcal{M} := \{\vartheta \in \mathcal{P}(X): \vartheta \ll \mu\}.$$

Since X is a Borel space, it is separable and metrizable, cf. Bertsekas and Shreve (1978, p. 118), so that the Borel σ -algebra $\mathcal{B}(X)$ is countably generated.[†] Therefore, by Cohn (2013, Proposition 3.4.5), the space $L^1(\mu)$ of all $(\mu$ -equivalence classes of) Borel measurable functions $g: X \rightarrow \mathbb{R}$ with $\int_X |g(x)| \mu(dx) < \infty$ is a separable Banach space and $\mathcal{P}(X)$ is separable and metrizable, cf. Bertsekas and Shreve (1978, Proposition 7.20). Moreover, the set $D(\mu)$ of all $g \in L^1(\mu)$ with $g \geq 0$ μ -a.s. and $\int_X g(x) \mu(dx) = 1$ is a closed subset of $L^1(\mu)$, so that it is a Borel space. Since, by Hölder's inequality and the Radon–Nikodym theorem, the map $D(\mu) \rightarrow \mathcal{P}(X)$, $g \mapsto g d\mu$ is continuous and injective, it follows from Kuratowski's theorem, cf. Bertsekas and Shreve (1978, Proposition 7.15), that $\mathcal{M}$ is Borel measurable and that the inverse $\mathcal{M} \rightarrow D(\mu)$, $\vartheta \mapsto \frac{d\vartheta}{d\mu}$ is Borel measurable.

Now, let $\alpha: D(\mu) \rightarrow [0, \infty]$ be a measurable function with $\alpha(g_0) = 0$ for some $g_0 \in D(\mu)$. Then, the cost function

$C: X \times \mathcal{P}(X) \rightarrow [0, \infty]$, given by

$$C(x, \vartheta) := \begin{cases} \alpha\left(\frac{d\vartheta}{d\mu}\right), & \text{if } \vartheta \in \mathcal{M}, \\ \infty, & \text{otherwise,} \end{cases}$$

is Borel measurable (and independent of $x \in X$). Then, $\kappa_0(x) := g_0 d\mu$ satisfies $C(x, \kappa_0(x)) = 0$ for all $x \in X$, and

$$\rho(f) = \sup_{\vartheta \ll \mu} \left(\int_X f(y) \vartheta(dy) - \alpha\left(\frac{d\vartheta}{d\mu}\right) \right) \quad \text{for all } f \in \mathcal{L}.$$

REMARK 4.2 Let $R: L^\infty(\mu) \rightarrow \mathbb{R}$ be a convex risk measure with $R(0) = 0$, cf. Föllmer and Schied (2016, Section 4.3), where $L^\infty(\mu)$ denotes the space of all (equivalence classes of) all measurable μ -essentially bounded functions $X \rightarrow \mathbb{R}$. Then, the map $\alpha: D(\mu) \rightarrow [0, \infty]$, given by

$$\begin{aligned}\alpha(g) &:= \sup_{f \in L^\infty(\mu)} \left(\int_X f(x) g(x) \mu(dx) - R(f) \right) \in [0, \infty] \\ &\text{for all } g \in D(\mu),\end{aligned}$$

is lower semicontinuous and thus measurable.[‡] If R is continuous from below, it follows that

$$R(f) = \sup_{\vartheta \ll \mu} \int_X f(y) \vartheta(dy) - \alpha\left(\frac{d\vartheta}{d\mu}\right) \quad \text{for all } f \in L^\infty(\mu),$$

cf. Föllmer and Schied (2016, Theorem 4.33). If, in addition, there exists some $\vartheta_0 \in \mathcal{M}$ with $\int_X f(y) \vartheta_0(dy) \leq R(f)$ for all $f \in L^\infty(\mu)$, then $\alpha(g_0) = 0$ for $g_0 := \frac{d\vartheta_0}{d\mu}$. Therefore, R is the restriction of a risk measure with weak optimal transport penalty to $L^\infty(\mu)$, i.e. $R(f) = \rho(f)$ for all $f \in L^\infty(\mu)$.

We conclude this subsection with two explicit examples that fall into the setup of this section.

EXAMPLE 4.3 (a) Let $\varphi: [0, \infty) \rightarrow \overline{\mathbb{R}}$ be convex with $\varphi(t) \in \mathbb{R}$ for all $t \in (0, \infty)$, $\varphi(1) = 0$, and $\varphi(0) = \lim_{t \downarrow 0} \varphi(t) \in [0, \infty]$. Then, φ is Borel measurable, and therefore the φ -divergence

$$\begin{aligned}\alpha: D(\mu) &\rightarrow [0, \infty], \\ g &\mapsto D_\varphi(g | \mu) := \int_X \varphi(g(x)) \mu(dx)\end{aligned}$$

is Borel measurable. Moreover, $\alpha(\mathbb{1}) = 0$, where $\mathbb{1}$ denotes the function that is constantly equal to one.

(b) Let $\lambda \in (0, 1)$ and $R(f) := \text{ES}_\mu^\lambda(f)$ is the λ -expected shortfall w.r.t. μ for all $f \in L^\infty(\mu)$. Then, by remark 4.2 and Föllmer and Schied (2016, Example 4.40), the map

$$\begin{aligned}\alpha: D(\mu) &\rightarrow [0, \infty], \\ g &\mapsto \alpha(g) := \begin{cases} 0, & \text{if } g \leq \lambda^{-1} \mu\text{-a.s.} \\ \infty, & \text{otherwise,} \end{cases}\end{aligned}$$

is Borel measurable. In particular, $\alpha(\mathbb{1}) = 0$ and therefore, $\rho(f) = \text{ES}_\mu^\lambda(f)$ for all $f \in L^\infty(\mu)$.

[†] Choose a metric consistent with the topology and a dense sequence $(x_n)_{n \in \mathbb{N}} \subset X$. Then, the system $\mathcal{G}$ consisting of all finite intersections of open balls with rational radii around x_n for $n \in \mathbb{N}$ forms a countable $\cap$ -stable generator of $\mathcal{B}(X)$.

[‡] In fact, by Hölder's inequality, the map $D(\mu) \rightarrow \mathbb{R}$, $g \mapsto \int_X f(x) g(x) \mu(dx) - R(f)$ is continuous for every $f \in L^\infty(\mu)$.

4.3. Wasserstein uncertainty

In this section, we consider penalties given in terms of Wasserstein distances. To that end, let $X = Y$ be a separable Banach space and $\mathcal{P}_p(X)$ denote the set of all probability measures $\nu \in \mathcal{P}(X)$ with finite moment $\int_X \|x\|^p \nu(dx) < \infty$ of order $p \in [1, \infty)$. We assume that $\mu \in \mathcal{P}_p(X)$ and consider the cost function

$$C(x, \vartheta) := \int_X \|y - x\|^p \vartheta(dy) \quad \text{for all } x \in X \text{ and } \vartheta \in \mathcal{P}(X).$$

Then, the cost C is of the form (7) with $c(x, y) = \|x - y\|^p$ and $\mathcal{M} = \mathcal{P}(X)$ and $\tau_x = \text{id}_X$ for all $x \in X$. Moreover, $\kappa_0(x) := \delta_x$ satisfies $C(x, \kappa_0(x)) = 0$ for all $x \in X$ and we end up with the risk measure

$$\begin{aligned} \rho(f) &= \sup_{\nu \in \mathcal{P}(X)} \left(\int_X f(y) \nu(dy) \right. \\ &\quad \left. - \inf_{\kappa \in \text{Ker}(\mu, \nu)} \int_X C((x, \kappa(x))) \mu(dx) \right) \\ &= \sup_{\nu \in \mathcal{P}_p(X)} \left(\int_X f(y) \nu(dy) - \mathcal{W}_p(\mu, \nu)^p \right) \\ &\quad \text{for all } f \in B(X), \end{aligned}$$

where $\mathcal{W}_p(\mu, \nu) := \inf_{\pi \in \text{Cpl}(\mu, \nu)} (\int_X \|y - x\|^p \pi(dx, dy))^{1/p}$ is the Wasserstein distance of order p between μ and ν . Different aspects of risk measures based on transport distances have recently been studied in Bartl *et al.* (2020, 2021) and Nendel and Sgarabottolo (2022). We also refer Bartl *et al.* (2021), Bartl and Wiesel (2023) and Fuhrmann *et al.* (2023) for dynamic versions of these risk measures and to Pflug and Wozabal (2007), Mohajerin Esfahani and Kuhn (2018), and Gao and Kleywegt (2023) for applications in the context of distributionally robust optimization. Since $\mathcal{P}_p(X)$ contains all Dirac measures, by proposition 3.4,

$$\begin{aligned} \rho(f) &= \int_X \sup_{y \in X} (f(y) - \|x - y\|^p) \bar{\mu}(dx) \\ &= \int_X \sup_{y \in X} (f(x + y) - \|y\|^p) \bar{\mu}(dx) \end{aligned}$$

for all $f \in \mathcal{L}$.

Now, let $f \in B(X)$ and assume that there exist constants $M \geq 0$ and $c \in [0, 1)$ such that

$$|f(x)| \leq M + c\|x\|^p \quad \text{for all } x \in X. \quad (11)$$

Since $\mu \in \mathcal{P}_p(X)$, it follows that $f \in \mathcal{L}^1(\mu)$. Using the convexity of the map $[0, \infty) \rightarrow [0, \infty)$, $v \mapsto v^p$, it follows that

$$\begin{aligned} |f(x + y)| &\leq M + \frac{c}{\lambda^{p-1}} \|x\|^p + \frac{c}{(1 - \lambda)^{p-1}} \|y\|^p \\ &\quad \text{for all } x, y \in X \text{ and } \lambda \in (0, 1). \end{aligned}$$

Now, let $\lambda \in (0, 1)$ such that $b := \frac{c}{(1 - \lambda)^{p-1}} < 1$ and $|\mu|_p := \int \|x\|^p \mu(dx)$. Then,

$$\begin{aligned} &\int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx) \\ &\leq M + \frac{c}{\lambda^{p-1}} |\mu|_p - (1 - b) \int_X \|y(x)\|^p \mu(dx) \end{aligned}$$

for all $y \in B(X, X)$. In particular, $f \in \mathcal{L}$ and, using corollary 3.7,

$$\rho(f) = \sup_{y \in L^p(\mu; X)} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dy), \quad (12)$$

where $L^p(\mu; X)$ denotes the space of all (μ -equivalence classes of) functions $y \in B(X, X)$ with $\int_X \|y(x)\|^p \mu(dx) < \infty$.

4.4. Drift uncertainty on the path space

Let $X = Y = C_0([0, 1])$ be the space of all continuous functions $x: [0, 1] \rightarrow \mathbb{R}^d$ with $x(0) = 0$, endowed with the topology induced by the supremum norm. Let $\mu \in \mathcal{P}(X)$ be the Wiener measure, and consider the cost function $c: X \times X \rightarrow [0, \infty]$, given by

$$c(x, y) := \begin{cases} \varphi(\|x - y\|_{H^1}^2), & \text{if } x - y \in H^1 \\ \infty, & \text{otherwise,} \end{cases}$$

with a nondecreasing function $\varphi: [0, \infty] \rightarrow [0, \infty]$. Here, $H^1 = H^1([0, 1])$ is the Cameron–Martin space, i.e. the space of all $u \in X$ with weak derivative $u' \in L^2 = L^2([0, 1])$, and $\|u\|_{H^1} := \|u'\|_{L^2}$ is the Cameron–Martin norm for $u \in H^1$. Note that, by definition of X , $u(0) = 0$ for all $u \in X$, so that the first Poincaré inequality

$$\|u\|_\infty \leq \int_0^1 |u'(t)| dt \leq \|u'\|_{L^2} = \|u\|_{H^1}$$

is valid for all $u \in H^1$. In particular, the cost function c is Borel measurable.

In this case, the cost function $C(x, \vartheta) := \int_X c(x, y) \vartheta(dy)$ is of the form (7) with $\mathcal{M} = \mathcal{P}(X)$ and $\tau_x = \text{id}_X$ for all $x \in X$. Moreover, $\kappa_0(x) := \delta_x$ satisfies $C(x, \kappa_0(x)) = 0$ for all $x \in X$. The corresponding risk measure is given by

$$\begin{aligned} \rho(f) &:= \sup_{\nu \in \mathcal{P}(X)} \left(\int_X f(y) \nu(dy) \right. \\ &\quad \left. - \inf_{\pi \in \text{Cpl}(\mu, \nu)} \int_{X \times X} c(x, y) \pi(dx, dy) \right) \end{aligned} \quad (13)$$

for all $f \in B(X)$. By proposition 3.4, it follows that

$$\begin{aligned} \rho(f) &= \int_X \sup_{u \in H^1} f(x + u) - \varphi(\|u\|_{H^1}^2) \bar{\mu}(dx) \\ &= \int_X \sup_{\theta \in L^2} \left(f\left(x + \int_0^\cdot \theta_s ds\right) \right. \\ &\quad \left. - \varphi\left(\int_0^1 |\theta_s|^2 ds\right) \right) \bar{\mu}(dx) \quad \text{for all } f \in \mathcal{L}. \end{aligned}$$

If $\varphi(v) = \frac{v}{2}$, by Föllmer (2022, Proposition 1) or Föllmer (2022, Proposition 2), it follows that

$$\begin{aligned} \rho(f) &\geq \sup_{v \in \mathcal{P}(X)} \left(\int_X f(x) v(dx) - H(v|\mu) \right) \\ &= \log \left(\int_X e^{f(x)} \mu(dx) \right) \end{aligned} \quad (14)$$

for all $f \in \mathcal{L}$, where $H(v|\mu)$ is the relative entropy of $v \in \mathcal{P}(X)$ with respect to μ . Note that, in general, equality in (14) only holds if the penalty term in (13) is replaced by the Cameron–Martin adapted Wasserstein distance, see Föllmer (2022, Corollary 2) for the details.

4.5. Martingale constraints

We consider the case, where $X = Y = \mathbb{R}^d$ and the ambiguity set is given by all measures that satisfy a martingale constraint with respect to the reference measure $\mu \in \mathcal{P}(\mathbb{R}^d)$. More precisely, we only consider probability measures $v \in \mathcal{P}(X)$ that admit a martingale coupling with the reference measure μ , i.e. a coupling $\pi \in \text{Cpl}(\mu, v)$ with

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x) \cdot \mathbb{1}_A(x) \pi(dx, dy) = 0 \quad \text{for all } A \in \mathcal{B}(X). \quad (15)$$

By Strassen (1965), the existence of a martingale coupling between two measures μ and v is equivalent to μ and v being in convex order. We consider the risk measure

$$\begin{aligned} \rho^{\text{Mart}}(f) &:= \sup_{v \in \mathcal{P}(\mathbb{R}^d)} \left(\int_{\mathbb{R}^d} f(y) v(dy) \right. \\ &\quad \left. - \inf_{\pi \in \text{Mart}(\mu, v)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(|y - x|) \pi(dx, dy) \right) \end{aligned} \quad (16)$$

for $f \in B(\mathbb{R}^d)$, where $c: [0, \infty) \rightarrow [0, \infty]$ is a Borel measurable cost function and $\text{Mart}(\mu, v)$ denotes the set of martingale couplings between μ and v .

Then, we can express this risk measure via a weak optimal transport penalty with cost function

$$C(x, \vartheta) = \begin{cases} \int_{\mathbb{R}^d} c(|y - x|) \vartheta(dy), & \text{if } \tau_x^{-1}(\vartheta) \in \mathcal{M}_0, \\ \infty, & \text{otherwise,} \end{cases}$$

with

$$\mathcal{M}_0 = \left\{ \vartheta \in \mathcal{P}(\mathbb{R}^d) : \int_{\mathbb{R}^d} x \vartheta(dx) = 0 \right\}$$

and $\tau_x(y) := x + y$ for all $x, y \in \mathbb{R}^d$, i.e. $\tau_x^{-1}(y) = y - x$ for all $x, y \in \mathbb{R}^d$.

In fact, let $v \in \mathcal{P}(\mathbb{R}^d)$, $\pi \in \text{Mart}(\mu, v)$, and $\kappa \in \text{Ker}(\mu, v)$ with $\pi(A \times B) = \int_A \kappa(x, B) \mu(dx)$ for all $A \in \mathcal{B}(X)$ and $B \in \mathcal{B}(Y)$, cf. remark 3.1. Then, by (15),

$$\begin{aligned} &\int_A \int_{\mathbb{R}^d} \tau_x^{-1}(y) \kappa(x, dy) \mu(dx) \\ &= \int_{\mathbb{R}^d \times \mathbb{R}^d} (y - x) \cdot \mathbb{1}_A(x) \pi(dx, dy) = 0 \\ &\quad \text{for all } A \in \mathcal{B}(X). \end{aligned}$$

Hence, $\tau_x^{-1}(\kappa(x)) \in \mathcal{M}_0$ for μ -almost all $x \in \mathbb{R}^d$.

On the other hand, every kernel $\kappa \in \text{Ker}(\mu, v)$ with $\tau_x^{-1}(\kappa(x)) \in \mathcal{M}_0$ gives rise to a coupling $\pi \in \text{Mart}(\mu, v)$ via $\pi(A \times B) := \int_A \kappa(x, B) \mu(dx)$ for all $A, B \in \mathcal{B}(\mathbb{R}^d)$. Hence,

$$\begin{aligned} &\inf_{\kappa \in \text{Ker}(\mu, v)} \int_{\mathbb{R}^d} C(x, \kappa(x)) \mu(dx) \\ &= \inf_{\pi \in \text{Mart}(\mu, v)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c(|y - x|) \pi(dx, dy), \end{aligned}$$

and therefore $\rho^{\text{Mart}} = \rho$.

Observe that the set $\mathcal{M}_0$ is defined by a single moment constraint, namely, a mean constraint. Hence, theorem 3.5 implies that

$$\begin{aligned} \text{ex } \mathcal{M}_0 &= \left\{ v \in \mathcal{P}(\mathbb{R}^d) : v = p\delta_x + (1 - p)\delta_{-\frac{p}{1-p}x}, \right. \\ &\quad \left. x \in X, p \in (0, 1) \right\}. \end{aligned}$$

Therefore, by proposition 3.6, the martingale constraint reduces to the following optimization

$$\begin{aligned} \rho(f) &= \int_{\mathbb{R}^d} \sup_{\substack{y \in X \\ p \in (0, 1)}} \left(p(f(x + y) - c(|y|)) \right. \\ &\quad \left. + (1 - p) \left(f\left(x - \frac{p}{1-p}y\right) - c\left(\frac{p}{1-p}|y|\right) \right) \right) \mu(dx). \end{aligned} \quad (17)$$

This means that the martingale constraint, which is an infinite-dimensional constraint, on the state space $\mathbb{R}^d$ reduces to a $d + 1$ -dimensional optimization problem. In section 5, we will use this representation of the martingale constraint to compute no-arbitrage bounds for prices of financial derivatives after quoted maturities.

4.6. Martingales on the Skorokhod space

Let $X = Y = \mathbb{R}^d$, and consider the space $\Omega := D = D([0, 1])$ of all càdlàg functions $[0, 1] \rightarrow \mathbb{R}^d$, endowed with the σ -algebra $\mathcal{F} := \sigma(\{p_t : t \in [0, 1]\})$ generated by the canonical process $(p_t)_{t \in [0, 1]}$. Let $\mu \in \mathcal{P}(\mathbb{R}^d)$ with $\int_{\mathbb{R}^d} |x|^2 \mu(dx) < \infty$. Then, we denote by $\text{Mart}_D^2(\mu)$ the set of all martingale measures η on $(\Omega, \mathcal{F})$ with $\eta \circ p_0^{-1} = \mu$ and $\int_{\Omega} |\omega_1|^2 \eta(d\omega) < \infty$. Consider the risk measure

$$\rho(f) := \sup_{\eta \in \text{Mart}_D^2(\mu)} \int_{\Omega} f(\omega_1) - \langle \omega \rangle_1 \eta(d\omega),$$

where $\langle \cdot \rangle$ denotes the quadratic variation. Then, by Itô's formula,

$$\int_{\Omega} \langle \omega \rangle_1 \eta(d\omega) = \int_{\Omega} |\omega_1|^2 \eta(d\omega) - \int_{\mathbb{R}^d} |x|^2 \mu(dx)$$

for all $\eta \in \text{Mart}_D^2(\mu)$. By Guo *et al.* (2017, Remark 2.2 (ii)), we thus find that

$$\rho(f) = \sup_{v \in \mathcal{P}(\mathbb{R}^d)} \int_{\mathbb{R}^d} f(y) v(dy) - \mathcal{W}_2^{\text{Mart}}(\mu, v)^2.$$

As in section 4.5, we can therefore apply theorem 2.1, and obtain that

$$\rho(f) = \int_{\mathbb{R}^d} \sup_{\theta \in \mathbb{R}^d} \sup_{p \in (0,1)} \left(pf \left(x + \frac{\theta}{p} \right) + (1-p)f \left(x - \frac{\theta}{1-p} \right) - \frac{|\theta|^2}{p(1-p)} \right) \mu(dx).$$

4.7. Matching quoted option prices on a given maturity

Let $X = Y = \mathbb{R}^d$ with $d \in \mathbb{N}$, and consider a two-period market consisting of d assets. We assume that the distribution of the assets today, i.e. at time $t = 0$, is known and given by a Dirac measure $\mu = \delta_{x_0}$ with $x_0 \in \mathbb{R}^d$. We assume that the discounted prices of the assets at time $t = 1$ are given by a random vector $x = (x_j)_{j=1,\dots,d}$, whose distribution is not known precisely.

Let $f_1, \dots, f_n: \mathbb{R} \rightarrow [0, \infty)$ be the payoffs of $n \in \mathbb{N}$ quoted options, e.g. call options $f_i(x) = (x_{j_i} - K_i)^+$ with $j_i \in \{1, \dots, d\}$ and $K_i \in [0, \infty)$ for $i = 1, \dots, n$. The market then gives a set of intervals $I_i = [b_i, a_i]$ with $b_i, a_i \in [0, \infty)$ representing the bid-ask spread for the option f_i for $i = 1, \dots, n$. If ν is the law of x , it should satisfy $\int_{\mathbb{R}^d} f_i(y) \nu(dy) \in [b_i, a_i]$, for all $i = 1, \dots, n$. This conditions can be represented as a set of $2n$ generalized moment constraints. In fact, for each i we consider the constraints

$$\int_{\mathbb{R}^d} f_i(y) \nu(dy) \leq a_i \quad \text{and} \quad \int_{\mathbb{R}^d} -f_i(y) \nu(dy) \leq -b_i.$$

Assuming, for simplicity, that the interest rate is zero, we impose absence of arbitrage by requiring that the measure ν satisfies the mean constraint $\int_{\mathbb{R}^d} y \nu(dy) = x_0$ at time $t = 1$, following the no-arbitrage argument proposed in Acciaio *et al.* (2016), where the authors provide a model-free version of the Fundamental Theorem of Asset Pricing in a market with quoted options. Intuitively, a measure that satisfies the above mean constraint does not give rise to an arbitrage opportunity since investing in the assets has an expected return zero.

We can then express the upper no-arbitrage price bound $\bar{V}(f)$ for a European option with payoff $f: \mathbb{R}^d \rightarrow \mathbb{R}$ and maturity $t = 1$ using the reference measure $\mu = \delta_{x_0}$ and the cost function $C(x, \vartheta) = \infty \cdot \mathbb{1}_{\mathcal{M}^c}(\vartheta)$ for $x \in \mathbb{R}^d$ and $\vartheta \in \mathcal{P}(\mathbb{R}^d)$ with $\mathcal{M}$ being the set of all probability measures that satisfy the $2n + 2$ constraints described above. We may then choose the kernel κ_0 arbitrarily among all kernels in $\text{Ker}(\mathbb{R}^d, \mathbb{R}^d)$ with $\kappa_0(x_0) \in \mathcal{M}$, and consider

$$\begin{aligned} \bar{V}(f) &:= \sup_{\nu \in \mathcal{M}} \int_{\mathbb{R}^d} f(y) \nu(dy) \\ &= \sup_{\nu \in \mathcal{P}(\mathbb{R}^d)} \left(\int_{\mathbb{R}^d} f(y) \nu(dy) - \inf_{\kappa \in \text{Ker}(\mu, \nu)} C(x_0, \kappa(x_0)) \right) \\ &= \rho(f). \end{aligned}$$

Here, the second and third equality directly follow from the choice of the reference measure $\mu = \delta_{x_0}$, which implies that $\kappa(x_0) = \nu \in \mathcal{M}$ for all $\kappa \in \text{Ker}(\mu, \nu)$ with $C(x_0, \kappa(x_0)) < \infty$.

Using the IRP of the set $\mathcal{M}$ and theorem 3.3, together with theorem 3.5, we can solve this problem by optimizing over

$2n + 3$ Dirac measures. This leads to

$$\rho(f) = \sup_{(p,y) \in \mathcal{M}} \sum_{i=1}^{2n+3} p_i f(y_i),$$

where $\mathcal{M} = \{(p, y) \in [0, 1]^{2n+3} \times (\mathbb{R}^d)^{2n+3} : \sum_{i=1}^{2n+3} p_i \delta_{y_i} \in \mathcal{M}\}$.

In a similar way, we can obtain the lower price bound by considering $\underline{V}(f) := -\rho(-f)$, see section 5.2 below.

5. Numerics

In this section, we complement our main result with additional approximation results, which are then applied in a series of numerical examples.

5.1. Wasserstein uncertainty without constraints

Let $X = Y$ be a separable Banach space. We start by considering the risk measure from section 4.3, i.e.

$$\rho(f) = \sup_{\nu \in \mathcal{P}(X)} \left(\int_X f(y) \nu(dy) - \mathcal{W}_p(\mu, \nu)^p \right) \quad \text{for } f \in \mathcal{B}(X) \quad (18)$$

with $p \geq 1$ and $\mu \in \mathcal{P}_p(X)$. Recall that the risk measure (18) corresponds to the weak transport cost $C(x, \vartheta) := \int_X |y - x|^p \vartheta(dy)$ for $x \in X$ and $\vartheta \in \mathcal{P}(X)$ and that, by (12), it can be expressed as

$$\rho(f) = \sup_{y \in L^p(\mu; X)} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx) \quad \text{for } f \in \mathcal{B}(X) \text{ with (11).}$$

The next proposition provides an approximation from below for this risk measure, based on an optimization over a set which is dense in $L^p(\mu; X)$.

PROPOSITION 5.1 *Let $f: X \rightarrow \mathbb{R}$ be continuous with (11), and let $(D_n)_{n \in \mathbb{N}}$ be a family of subsets of $L^p(\mu; X)$ with $D^n \subset D^{n+1}$ for all $n \in \mathbb{N}$. If $D := \bigcup_{n \in \mathbb{N}} D_n$ is dense in $L^p(\mu; X)$, then the risk measure (18) can be approximated from below in the following sense:*

$$\sup_{y \in D^n} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx) \nearrow \rho(f) \quad \text{as } n \rightarrow \infty.$$

Proof Since $D \subset L^p(\mu; X)$, it follows that

$$\rho(f) \geq \sup_{y \in D} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx).$$

To prove the other inequality, let $\varepsilon > 0$ and $y_0 \in L^p(\mu; X)$ with

$$\sup_{y \in L^p(\mu; X)} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx)$$

$$\leq \int_X (f(x + y_0(x)) - \|y_0(x)\|^p) \mu(dx) + \frac{\varepsilon}{2}.$$

For $y \in L^p(\mu; X)$, consider the probability measure $\mu_y := \mu \circ (x \mapsto x + y(x))^{-1} \in \mathcal{P}_p(X)$. Then, using the comonotone coupling,

$$\mathcal{W}_p(\mu_{y_0}, \mu_y) \leq \|y_0 - y\|_{L^p(\mu; X)} \quad \text{for all } y \in L^p(\mu; X).$$

Since D is dense in $L^p(\mu; X)$ and the Wasserstein distance metrizes the weak topology on $\mathcal{P}_p(X)$, cf. Villani (2009, Definition 8.8 and Theorem 6.9), there exists some $y_D \in L^p(\mu; X)$ with

$$\begin{aligned} & \int_X (f(x + y_0(x)) - \|y_0(x)\|^p) \mu(dx) \\ & \leq \int_X (f(x + y_D(x)) - \|y_D(x)\|^p) \mu(dx) + \frac{\varepsilon}{2}. \end{aligned}$$

Altogether, we find that

$$\begin{aligned} & \sup_{y \in L^p(\mu; X)} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx) \\ & \leq \int_X (f(x + y_D(x)) - \|y_D(x)\|^p) \mu(dx) + \varepsilon. \end{aligned}$$

Letting $\varepsilon \downarrow 0$ and taking the supremum over all $y \in D$, we obtain that

$$\rho(f) = \sup_{y \in D} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx). \quad (19)$$

Since $D^n \subseteq D^{n+1}$ for all $n \in \mathbb{N}$,

$$\begin{aligned} & \sup_{y \in D} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx) \\ & = \sup_{n \in \mathbb{N}} \sup_{y \in D_n} \int_X (f(x + y(x)) - \|y(x)\|^p) \mu(dx), \end{aligned}$$

and the claim follows from equation (19). $\blacksquare$

The previous proposition provides a constructive way of approximating the risk measure (18): given an increasing sequence of subsets of $L_p(\mu; X)$, whose union is dense in $L^p(\mu; X)$, we can approximate the risk measure from below by solving a variational problem on the increasing sequence of subsets. From a practical point of view, the previous result allows to approximate the C -transform of the loss function by looking at an optimizing vector field instead of solving a pointwise optimization.

Following closely the approach in Nendel and Sgarabottolo (2022, Section 3), we can apply this result in the case $X = \mathbb{R}^d$ with $d \in \mathbb{N}$ using the numerical framework of neural networks. For the sake of a self-contained exposition, we provide a full description of the framework and restate the approximation result for this explicit choice of increasing

subsets. We denote by $\mathfrak{N}_d^{m,n}$ the set of fully connected feed-forward neural networks from $\mathbb{R}^d$ to $\mathbb{R}^d$ with m hidden layers and n neurons per layer, i.e. the set of functions

$$x \mapsto A_m \circ \psi \circ A_{m-1} \circ \dots \circ \psi \circ A_0(x),$$

where A_i are affine transformations of the form $A_i = M_i x + b_i$ with a matrix M_i and a vector b_i , $\psi: \mathbb{R} \rightarrow \mathbb{R}$ is a nonlinear function, called the *activation function*, and the composition with the activation function is to be understood component-wise, i.e.

$$\psi((x_1, \dots, x_d)) = (\psi(x_1), \dots, \psi(x_d)).$$

For a neural network in $\mathfrak{N}_d^{m,n}$, A_0 is a function from $\mathbb{R}^d$ to $\mathbb{R}^n$, $A_1, \dots, A_{m-1}$ are functions from $\mathbb{R}^n$ to $\mathbb{R}^n$, and A_m is a function from $\mathbb{R}^n$ to $\mathbb{R}^d$. The fact that $\mathfrak{N}_d^{m,n}$ is dense in $L^p(\mu; \mathbb{R}^d)$ for suitable activation functions allows us to state the following corollary. Its proof relies on classical universal approximation results, cf. Hornik *et al.* (1989) and Hornik (1991), and follows closely the one of Nendel and Sgarabottolo (2022, Corollary 3.3).

COROLLARY 5.2 *For every $m \in \mathbb{N}$ and every nonconstant Lipschitz continuous activation function ψ ,*

$$\sup_{y \in \mathfrak{N}_d^{m,n}} \int_{\mathbb{R}^d} (f(x + y(x)) - |y(x)|^p) \mu(dx) \nearrow \rho(f) \quad \text{as } n \rightarrow \infty$$

for all continuous functions $f: \mathbb{R}^d \rightarrow \mathbb{R}$ with (11).

Proof Since the activation function ψ is Lipschitz continuous, the functions in $\mathfrak{N}_d^{m,n}$ are also Lipschitz continuous as a composition of Lipschitz continuous functions, so that $\mathfrak{N}_d^{m,n} \subset L^p(\mu; X)$. For $m = 1$, the convergence follows directly from proposition 5.1 and Hornik (1991, Theorem 1). Moreover, $\mathfrak{N}_d^{1,n} \subseteq \mathfrak{N}_d^{m,n}$, which proves the approximation for all $m \in \mathbb{N}$. $\blacksquare$

In this framework, the optimization in the space of neural networks can be performed via gradient descent methods minimizing the loss function

$$L(y) = \int_{\mathbb{R}^d} (f(x + y(x)) - |y(x)|^p) \mu(dx),$$

where the integral is computed numerically, using either quadrature formulas or a Monte Carlo simulation. Note that, when computing the integral with a Monte Carlo simulation, new samples are drawn at each step of the training, so that gradient descent automatically results in a stochastic gradient descent algorithm.

5.1.1. A two-dimensional example with Wasserstein-2 penalization. We provide an example for this framework recalling the earthquake example from Nendel and Sgarabottolo (2022). We consider a one-period model, where $X = \mathbb{R}^2$ and the reference measure μ describes an a priori estimate for the location of the epicenter of an earthquake. Suppose an insurance company, selling coverage for the damages caused

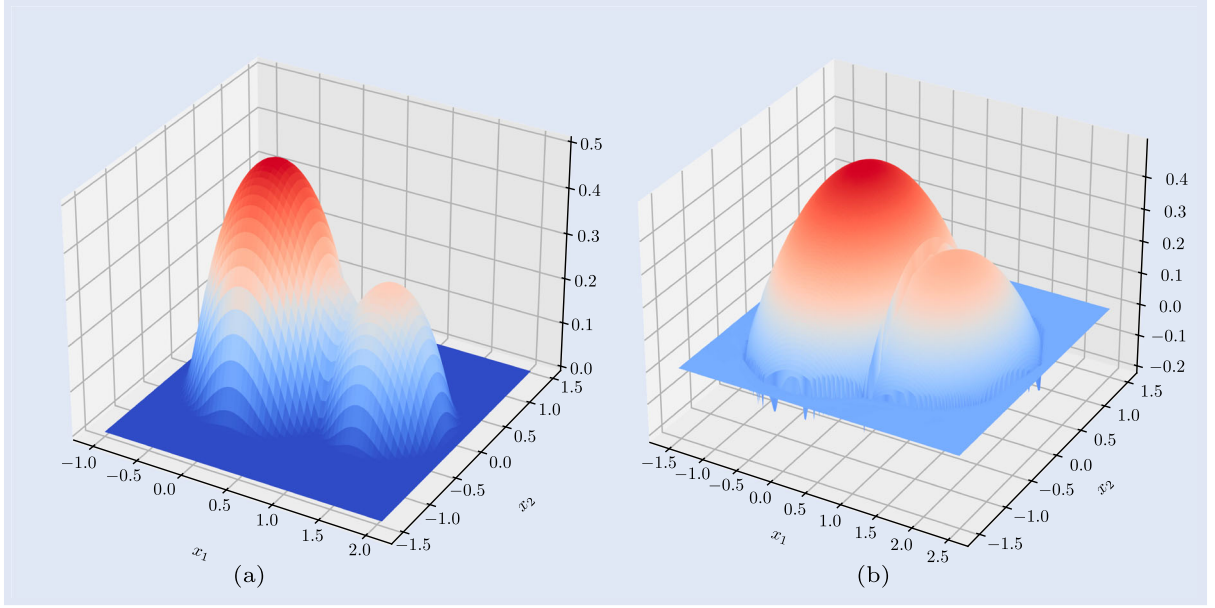


Figure 1. On the left, loss function for the example with Wasserstein uncertainty without constraints. On the right, corresponding C -transform. (a) Loss function f (b) C -transform of f .

by earthquakes, estimates a loss function that depends on the location of the epicenter of the earthquake and computes the worst-case loss penalizing deviations from the reference model with the Wasserstein-2 distance. We thus consider the risk measure (18) with $p = 2$, that is

$$\rho(f) = \sup_{\nu \in \mathcal{P}_2(\mathbb{R}^2)} \left(\int_{\mathbb{R}^2} f(x) \nu(dx) - \mathcal{W}_2(\mu, \nu)^2 \right).$$

Using theorem 2.1, one may compute the risk measure ρ by estimating the function f^C and computing its expectation with respect to the reference measure μ , e.g. via a Monte Carlo simulation. We work with the loss function depicted in figure 1(a), which models the dependence of the losses on the density of the buildings in a city, or on the distance from two city centers with different exposures. As a reference measure μ , we choose the law of a two-dimensional Gaussian random variable with mean located at $(0.75, 0.25)$ and identity matrix as covariance matrix. We estimate this risk measure by approximating f^C with the neural network approach described above.

We use a neural network consisting of 4 hidden layers and 20 neurons per layer with ReLU activation function, i.e. $\psi(x) = \max\{x, 0\}$. The optimization of the neural network is obtained using the Adam optimizer, cf. Kingma and Ba (2014), with a learning rate of 0.001 and a batch size of 100 samples, which means that at each iteration only 100 sample points are used to compute the Monte Carlo.† Figure 2 shows the training phase of the risk measure with the neural network approach and figure 1(b) depicts the resulting C -transform after 10 000 training epochs. In order to obtain a numerical value for $\rho(f)$, one only has to perform

a final Monte Carlo simulation to integrate the resulting C -transform f^C with respect to the reference measure μ , and this last evaluation can be performed at a low computational cost with a large sample, e.g. one million points. Note that, in order to choose the number of layers and the number of neurons per layer, we studied the training cycle of the loss function, cf. figure 2, across different network configurations. In our framework assessing the performance of different networks is simple, since we always provide an approximation of the risk measure from below. Hence, the network that allows to reach the highest value in the training phase is chosen to display the results. It has to be noted that many different configurations can actually lead to the same results. The rule of thumb in this situation is to choose an architecture which is not unnecessarily complicated. Changing payoff function or cost function might also require to select a different network architecture in order to perform the evaluation, therefore, an expert judgement is always needed in this phase.

5.2. Martingale constraint: pricing after quoted maturities

In this section, we apply the framework from section 4.5 to the pricing of derivatives in a model-free setting. We assume that $X = Y = \mathbb{R}^d$ with $d \in \mathbb{N}$, and consider an arbitrage-free market with d assets and risk-free rate zero. Suppose that there are many options quoted on a certain maturity $T > 0$, and we therefore know the distribution of the assets at time T , and that there are no more quoted options after this maturity. If we want to price a European derivative with maturity $T + t$, we then have no information for the calibration of a model at that maturity. Using, however, a model-free approach, we can compute bounds on the option price looking at all probability measures which dominate the distribution of the underlying at time T in convex order. In fact, even without making rigorous assumptions on the market model, we can assume that the

† All numerical analyses were performed on a laptop equipped with a 4-Core 3.48GHz CPU and 16GB RAM. The examples were implemented in Python 3.9 and the neural network architecture was handled using the package PyTorch (version 1.11.0). Source codes are available at https://github.com/sgarale/risk_meas_wot.

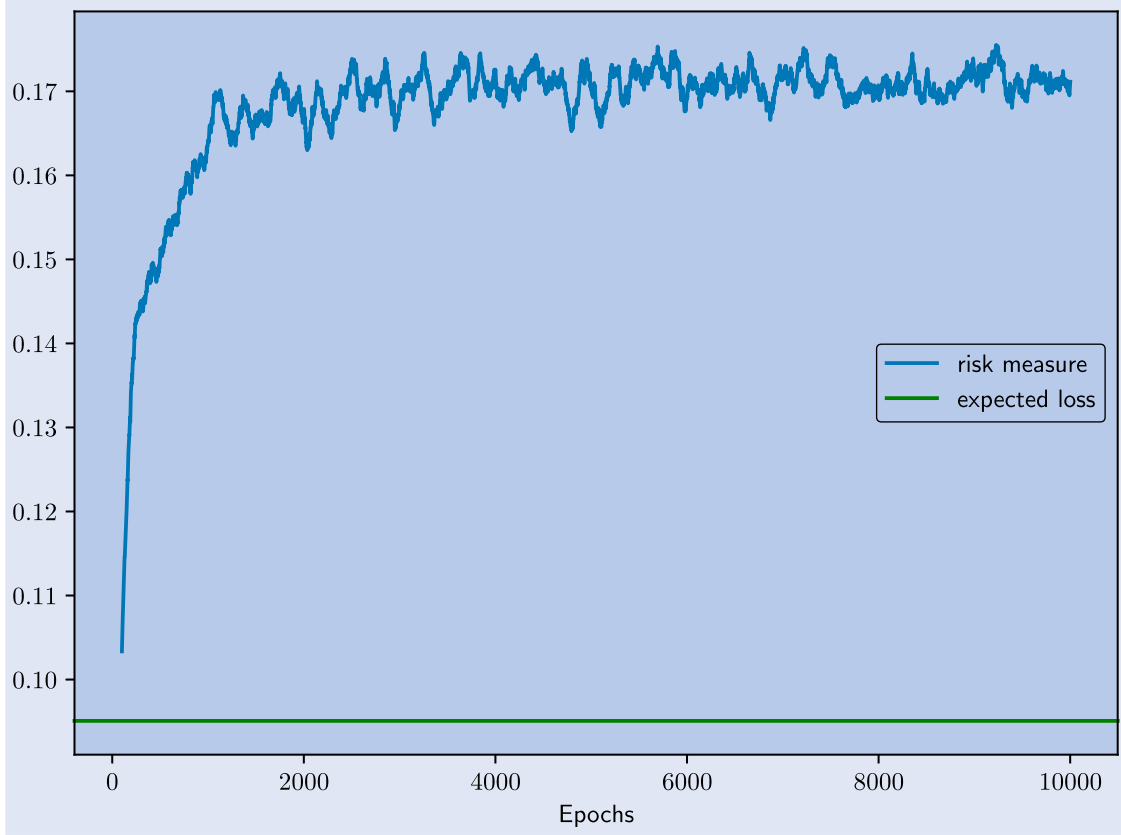


Figure 2. Training of the risk measure for the two-dimensional example with Wasserstein penalization. The values plotted are moving averages with rolling window of 100 epochs.

absence of arbitrage implies that (discounted) asset processes are martingales.

As a best guess for the distribution at time $T + t$, we choose the distribution μ of the assets at time T , and we penalize deviations from μ by means of a martingale optimal transport cost as discussed in section 4.5.

In order to take into account the length of the time step t , we introduce a parameter that controls the level of uncertainty by a rescaling of the penalty term, considering the cost $c_t(v) := tc(\frac{v}{\sqrt{t}})$ for $t > 0$ and $v \geq 0$. Roughly speaking, the assumption underlying this rescaling is that the uncertainty in the distribution grows proportionally to the square root of time.

We denote by $\bar{V}_{T+t}(f)$ and $\underline{V}_{T+t}(f)$ the upper and lower bound for the option price at time $T + t$, respectively, i.e.

$$\begin{aligned} \bar{V}_{T+t}(f) &= \rho_t^{\text{Mart}}(f) \\ &:= \sup_{\nu \in \mathcal{P}(\mathbb{R}^d)} \left(\int_{\mathbb{R}^d} f(y) \nu(dy) \right. \\ &\quad \left. - \inf_{\pi \in \text{Mart}(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c_t(|y - x|) \pi(dx, dy) \right). \end{aligned}$$

Note that $\rho_t^{\text{Mart}}(f)$ corresponds to equation (16) with an additional time scaling parameter.

Using the same arguments, it is natural to define the lower bound for the option price considering the infimum over probability measures that dominate μ in convex order and penalize

them with the same cost. This leads to

$$\begin{aligned} \underline{V}_{T+t}(f) &:= \inf_{\nu \in \mathcal{P}(\mathbb{R}^d)} \left(\int_{\mathbb{R}^d} f(y) \nu(dy) \right. \\ &\quad \left. + \inf_{\pi \in \text{Mart}(\mu, \nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} c_t(|y - x|) \pi(dx, dy) \right) \\ &= -\rho_t^{\text{Mart}}(-f). \end{aligned}$$

Numerically, we will solve the reduced problem (17) with $c(v) = |v|^p$ and with the additional time parameter in the cost function. As in the previous example, we solve this problem using proposition 3.6, optimizing outside of the integral and reducing the optimization to the space of neural networks with given width and depth, i.e. we consider the optimization

$$\begin{aligned} &\sup_{(y, \nu) \in \mathfrak{N}_{d, d+1}^{m, n}} \int_X \left(\varphi(\nu(x)) (f(x + y(x)) - c(|y(x)|)) \right. \\ &\quad \left. + (1 - \varphi(\nu(x))) \left[f\left(x - \frac{\varphi(\nu(x))}{1 - \varphi(\nu(x))} y(x)\right) \right. \right. \\ &\quad \left. \left. - c_t\left(\frac{\varphi(\nu(x))}{1 - \varphi(\nu(x))} |y(x)|\right) \right] \right) \mu(dx), \end{aligned}$$

where $(y, \nu) \in \mathfrak{N}_{d, d+1}^{m, n}$, indicates a neural network from $\mathbb{R}^d$ to $\mathbb{R}^{d+1}$, and we denote its components by $y: \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\nu: \mathbb{R}^d \rightarrow \mathbb{R}$. Moreover, we use the transform $\varphi(v) = \frac{1}{1+e^{-v}}$ to map real numbers to probabilities.

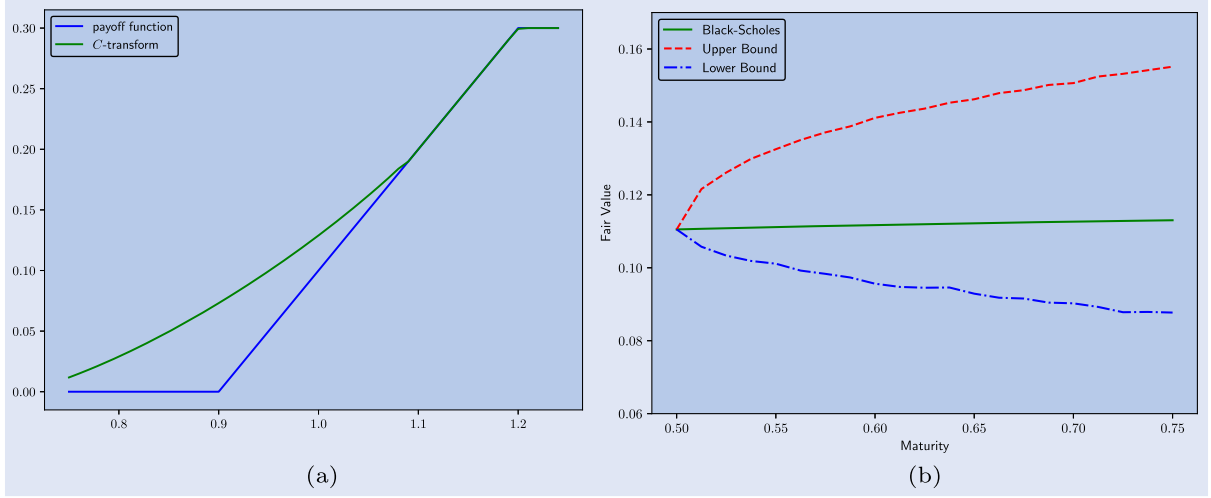


Figure 3. On the left, payoff function and C -transform for the upper bound price of the bull spread option 1 month after the quoted maturity. On the right, upper and lower bounds for the price of the option at different times after the quoted maturity. Quoted maturity: $T = 0.5$. Reference model: Black–Scholes model with volatility $\sigma = 0.2$. (a) Payoff and its C -transform (b) Upper and lower price bounds.

5.2.1. Bull spread option. We consider a bull call spread option for different maturities which follow a quoted maturity $T > 0$. A bull call spread option is a European option with payoff given by the sum of a long call option with strike $K_1 > 0$ and a short call option with strike $K_2 > K_1$. This leads to the payoff function

$$f(x) = (x - K_1)^+ - (x - K_2)^+.$$

As a reference model, we consider the log-normal distribution corresponding to the marginal at time $T = 0.5$ of a Black–Scholes model with volatility $\sigma = 0.2$. We set the initial value of the asset to one and the strikes to $K_1 = 0.9$ and $K_2 = 1.2$.

We work with the cost function $c(v) = |v|^3$, which corresponds to a Wasserstein-3 penalization and compute the upper bound $\bar{V}_{T+t}(f)$ and the lower bound $\underline{V}_{T+t}(f)$ for the arbitrage-free price of the option at different maturities after the quoted maturity T . We compare these bounds with the price given by the reference model, i.e. the Black–Scholes price. The output is displayed in figure 3(b). In figure 3(a), we depict the payoff function and the resulting C -transform for a specific maturity.

5.2.2. Max call option on two assets. We extend the previous example to a market with two assets. We compute the upper bound for the price of a max call option, which is the European option defined by the payoff

$$f(x_1, x_2) = (\max\{x_1, x_2\} - K)^+,$$

with $K > 0$. As a reference measure, we consider the marginal at time $T = 1$ of a diffusive model composed by two assets whose dynamics under the equivalent martingale measure $\mathbb{Q}$ is given by the SDE

$$\begin{cases} dX_t = \sigma dB_t, \\ X_0 = x_0, \end{cases}$$

with $X_t = \begin{pmatrix} X_t^1 \\ X_t^2 \end{pmatrix}$, $x_0 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $B_t = \begin{pmatrix} B_t^1 \\ B_t^2 \end{pmatrix}$,

$$\sigma = \begin{pmatrix} 0.30 & 0 \\ 0.05 & 0.20 \end{pmatrix},$$

where $(B_t)_t$ is a standard two-dimensional Brownian motion under $\mathbb{Q}$, see Karatzas and Shreve (1998, Chapter 1). We fix $t = 1/12$ and penalize deviations from the reference measure with a Wasserstein-3 penalty. As before, we can employ the previously described neural network approach to compute the upper bound for the price of the max call option with strike at the money $K = 1$. Figure 4 displays the payoff function and its C -transform.

5.2.3. Higher dimensions. We can also work in markets with a higher number of assets. As a matter of fact, the training of the neural network is based on sampling from the reference measure and therefore avoids the *curse of dimensionality*, which arises in methods based on the discretization of the underlying space. As the number of assets in the market increases, the network reproducing the measurable selection of the optimizers for the C -transform of the payoff function becomes larger, but this does not dramatically affect the performance of the method. In figure 5, we display the result of the evaluation of the upper bound for the price of several options as the number of underlying assets increases, compared with their evaluation time (in seconds). For each value, we train the neural network for 10^4 epochs. Again, the reference model is a diffusive model in which each asset has volatility 0.2 and is driven by a standard Brownian motion. Moreover, we assume that the assets start at time zero from the value 1. Clearly, as in the previous section, it is possible to impose any correlation structure between the assets. We assume that the quoted maturity is $T = 1$ (one year) and we compute the upper bound for the price at time $T + t$ with $t = 1/12$, i.e. one month after the quoted maturity.

We consider four different options, which have the following payoffs:

$$\text{max call: } f(x_1, \dots, x_d) = (\max\{x_1, \dots, x_d\} - K)^+$$

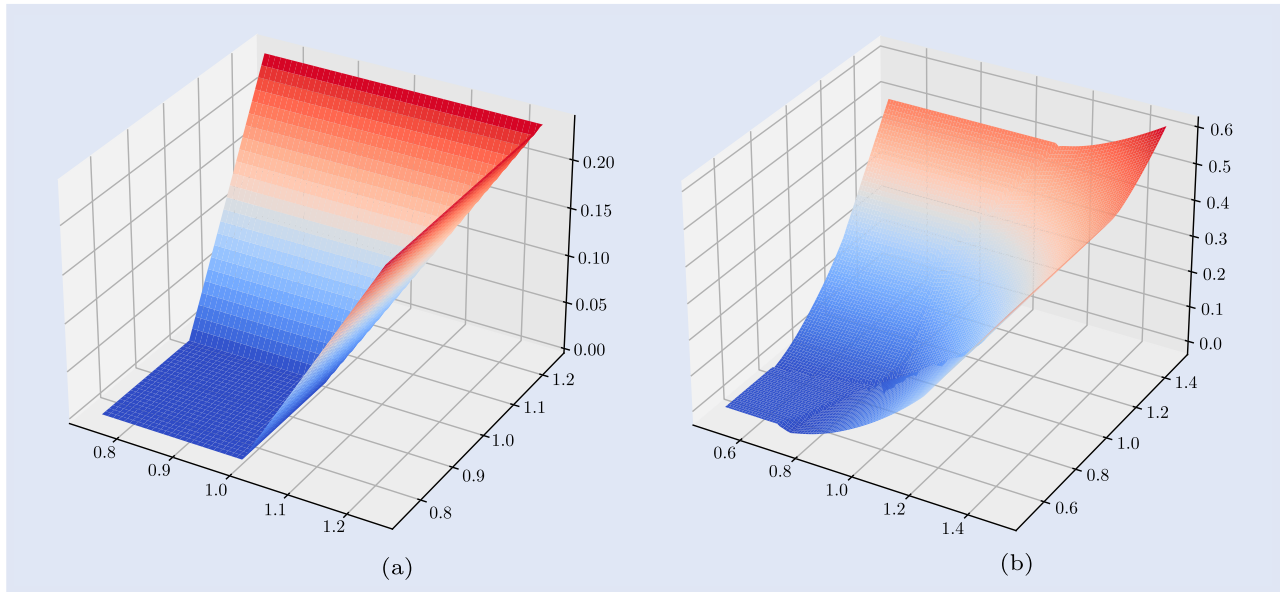


Figure 4. Payoff of the max call option on two assets and corresponding C -transform for the upper bound price with maturity 1 year and 1 month. Strike level $K = 1$. (a) Payoff function (b) C -transform.

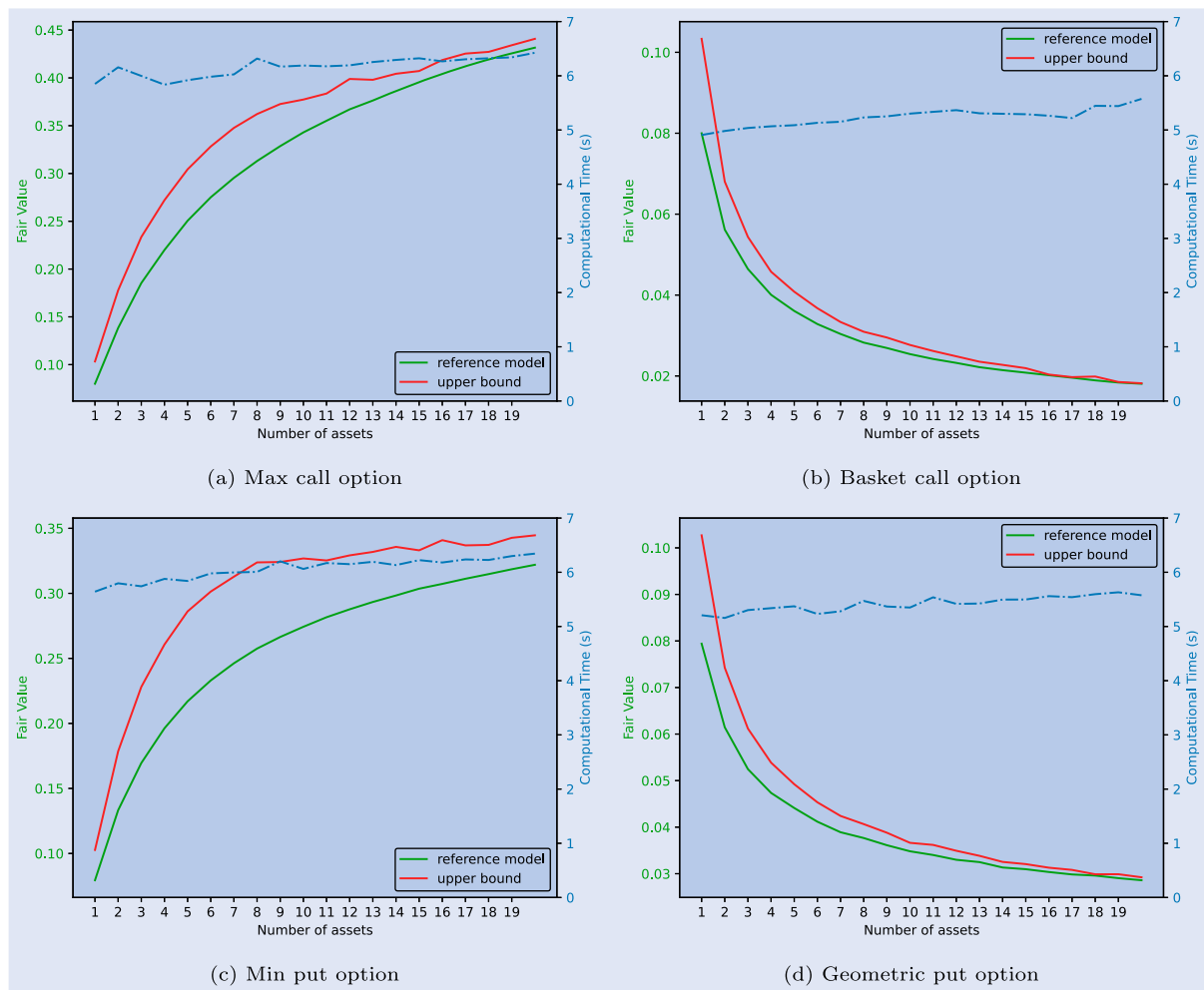


Figure 5. Option prices and evaluation time in seconds as the number of assets in the market increases. Strike of the options $K = 1$. (a) Max call option (b) Basket call option (c) Min put option (d) Geometric put option.

$$\text{basket call: } f(x_1, \dots, x_d) = \left(\frac{1}{d} \sum_{i=1}^d x_i - K \right)^+$$

$$\text{min put: } f(x_1, \dots, x_d) = (K - \min\{x_1, \dots, x_d\})^+$$

$$\text{geometric put: } f(x_1, \dots, x_d) = \left(K - \left(\prod_{i=1}^d x_i \right)^{\frac{1}{d}} \right)^+$$

These options depend on a single strike $K > 0$, which, in the numerical examples, is set to be at the money $K = 1$.

Acknowledgments

The authors thank Giacomo Scandolo and three anonymous referees for helpful comments related to this work.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was financially supported through the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation)—SFB 1283/2 2021—317210226 is gratefully acknowledged.

References

- Acciaio, B., Beiglbock, M., Penkner, F. and Schachermayer, W., A model-free version of the fundamental theorem of asset pricing and the super-replication theorem. *Math. Finance*, 2016, **26**, 233–251.
- Alibert, J.J., Bouchitté, G. and Champion, T., A new class of costs for optimal transport planning. *Eur. J. Appl. Math.*, 2019, **30**, 1229–1263.
- Backhoff, J., Beiglbock, M., Lin, Y. and Zolashko, A., Causal transport in discrete time and applications. *SIAM J. Optim.*, 2017, **27**, 2528–2562.
- Backhoff-Veraguas, J., Bartl, D., Beiglbock, M. and Eder, M., Adapted Wasserstein distances and stability in mathematical finance. *Finance Stoch.*, 2020, **24**, 601–632.
- Backhoff-Veraguas, J., Beiglbock, M. and Pammer, G., Existence, duality, and cyclical monotonicity for weak transport costs. *Calc. Var. Partial Differ. Equ.*, 2019, **58**, 203.
- Backhoff-Veraguas, J. and Pammer, G., Applications of weak transport theory. *Bernoulli*, 2022, **28**, 370–394.
- Bartl, D., Drapeau, S., Oblój, J. and Wiesel, J., Sensitivity analysis of Wasserstein distributionally robust optimization problems. *Proc. R. Soc. A*, 2021, **477**, 20210176.
- Bartl, D., Drapeau, S. and Tangpi, L., Computational aspects of robust optimized certainty equivalents and option pricing. *Math. Finance*, 2020, **30**, 287–309.
- Bartl, D., Eckstein, S. and Kupper, M., Limits of random walks with distributionally robust transition probabilities. *Electron. Commun. Probab.*, 2021, **26**, 28.
- Bartl, D., Neufeld, A. and Park, K., Sensitivity of robust optimization problems under drift and volatility uncertainty. Preprint, 2023. [arXiv:2311.11248](https://arxiv.org/abs/2311.11248).
- Bartl, D. and Wiesel, J., Sensitivity of multiperiod optimization problems with respect to the adapted Wasserstein distance. *SIAM J. Financial Math.*, 2023, **14**, 704–720.
- Beiglbock, M., Henry-Labordère, P. and Penkner, F., Model-independent bounds for option prices—A mass transport approach. *Finance Stoch.*, 2013, **17**, 477–501.
- Bernard, C., Pesenti, S.M. and Vanduffel, S., Robust distortion risk measures. Preprint, 2022. [arXiv:2205.08850](https://arxiv.org/abs/2205.08850).
- Bertsekas, D.P. and Shreve, S.E., *Stochastic Optimal Control: The Discrete Time Case*, Mathematics in Science and Engineering, Vol. 139, 1978 (Academic Press [Harcourt Brace Jovanovich]: New York).
- Blanchet, J. and Murthy, K., Quantifying distributional model risk via optimal transport. *Math. Oper. Res.*, 2019, **44**, 565–600.
- Cohn, D.L., *Measure Theory*, 2nd ed., Birkhäuser Advanced Texts: Basel Textbooks, 2013 (Birkhäuser/Springer: New York).
- De Gennaro Aquino, L. and Eckstein, S., Minmax methods for optimal transport and beyond: Regularization, approximation and numerics. *Adv. Neural. Inf. Process. Syst.*, 2020, **33**, 13818–13830.
- Eckstein, S. and Kupper, M., Computation of optimal transport and related hedging problems via penalization and neural networks. *Appl. Math. Optim.*, 2021, **83**, 639–667.
- Eckstein, S., Kupper, M. and Pohl, M., Robust risk aggregation with neural networks. *Math. Finance*, 2020, **30**, 1229–1272.
- Föllmer, H. and Schied, A., *Stochastic Finance: An Introduction in Discrete Time*, De Gruyter Graduate. Fourth revised and extended ed., 2016 (De Gruyter: Berlin).
- Föllmer, H., Optimal couplings on Wiener space and an extension of Talagrand’s transport inequality. In *Stochastic Analysis, Filtering, and Stochastic Optimization*, pp. 147–175, 2022 (Springer: Cham).
- Fuhrmann, S., Kupper, M. and Nendel, M., Wasserstein perturbations of Markovian transition semigroups. *Ann. Inst. Henri Poincaré Probab. Stat.*, 2023, **59**, 904–932.
- Gao, R. and Kleywegt, A., Distributionally robust stochastic optimization with Wasserstein distance. *Math. Oper. Res.*, 2023, **48**, 603–655.
- Gozlan, N., Roberto, C., Samson, P.M. and Tetali, P., Kantorovich duality for general transport costs and applications. *J. Funct. Anal.*, 2017, **273**, 3327–3405.
- Guo, G., Tan, X. and Touzi, N., Tightness and duality of martingale transport on the Skorokhod space. *Stoch. Process. Appl.*, 2017, **127**, 927–956.
- Henry-Labordère, P., (Martingale) optimal transport and anomaly detection with neural networks: A primal-dual algorithm. Preprint, 2019. [arXiv:1904.04546](https://arxiv.org/abs/1904.04546).
- Hornik, K., Approximation capabilities of multilayer feedforward networks. *Neural. Netw.*, 1991, **4**, 251–257.
- Hornik, K., Stinchcombe, M. and White, H., Multilayer feedforward networks are universal approximators. *Neural. Netw.*, 1989, **2**, 359–366.
- Karatzas, I. and Shreve, S.E., *Methods of Mathematical Finance*, Applications of Mathematics (New York), Vol. 39, 1998 (Springer-Verlag: New York).
- Kingma, D.P. and Ba, J., Adam: A method for stochastic optimization. In *International Conference on Learning Representations*. Preprint, 2014. [arXiv:1412.6980](https://arxiv.org/abs/1412.6980).
- Mohajerin Esfahani, P. and Kuhn, D., Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Math. Program.*, 2018, **171**, 115–166.
- Nendel, M. and Sgarabottolo, A., A parametric approach to the estimation of convex risk functionals based on Wasserstein distance. Preprint, 2022. [arXiv:2210.14340](https://arxiv.org/abs/2210.14340).
- Oblój, J. and Wiesel, J., Robust estimation of superhedging prices. *Ann. Statist.*, 2021, **49**, 508–530.
- Pesenti, S., Wang, Q. and Wang, R., Optimizing distortion riskmetrics with distributional uncertainty. Preprint, 2020. [arXiv:2011.04889](https://arxiv.org/abs/2011.04889).

- Pflug, G. and Wozabal, D., Ambiguity in portfolio selection. *Quant. Finance*, 2007, **7**, 435–442.
- Strassen, V., The existence of probability measures with given marginals. *Ann. Math. Statist.*, 1965, **36**, 423–439.
- Tan, X. and Touzi, N., Optimal transportation under controlled stochastic dynamics. *Ann. Probab.*, 2013, **41**, 3201–3240.
- Topsøe, F., *Topology and Measure*, Lecture Notes in Mathematics, Vol. 133, 1970 (Springer-Verlag: Berlin).
- Villani, C., *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences], Vol. 338, 2009 (Springer-Verlag: Berlin).
- von Weizsäcker, H. and Winkler, G., Integral representation in the set of solutions of a generalized moment problem. *Math. Ann.*, 1979, **246**, 23–32.
- Williams, D., *Probability with Martingales*, Cambridge Mathematical Textbooks, 1991 (Cambridge University Press: Cambridge).
- Winkler, G., Extreme points of moment sets. *Math. Oper. Res.*, 1988, **13**, 581–587.
- Zhao, C. and Guan, Y., Data-driven risk-averse stochastic optimization with Wasserstein metric. *Oper. Res. Lett.*, 2018, **46**, 262–267.